








# Modelling Financial Markets

## Chapter 16: Digital Assets and DeFi



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# Digital Assets: What Is Different and How to Measure It

## Digital Assets, Pricing and Liquidity

- The question: do cryptos, stablecoins and DeFi pools behave like familiar assets?
  - which measurements must change?
- Part I: the crypto market, stylised facts of returns, crypto indices (CRIX), stablecoins
- Part II: DeFi and automated market makers, spot ETFs and crypto equities
  - DeFi: decentralised finance
  - spot ETFs: exchange-traded funds that hold the coin
  - Cryptos as an asset class, a case study, one pass of the research loop assisted by artificial intelligence (AI)
- Detailed examples and advanced topics are in Further Topics, after the Conclusions

## What you should be able to do

- compute the statistics of a 7-day market on the right calendar
  - with intervals for its tail index, volatility and correlations
- compute a market-value index with a divisor
  - estimate a stablecoin peg as a threshold autoregression (threshold AR)
- derive the price of a swap in a constant-product pool and interpret providing liquidity as selling volatility
- interpret ETF tracking, crypto betas and asset-class claims
  - with tests that respect closing times and unknown break dates
- Further Topics and Appendix: reading for familiarity

# Blockchains, Coins and Tokens




Bust of Satoshi Nakamoto, Graphisoft Park, Budapest

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- **Blockchain** Nakamoto (2008): a shared ledger of blocks, each carrying the hash of the previous one
  - Secured by proof of work (Bitcoin) or proof of stake (Ethereum since 15 September 2022, ethereum.org: The Merge)
- Bitcoin issuance halves over time up to 21 million coins Bitcoin.org FAQ
  - 20.09 million existed on 18 Sep 2026
- **Coins** are native assets of a blockchain
  - **tokens** (stablecoins, governance tokens, tokenised funds) are issued by **smart contracts** Buterin (2014)
  - DeFi is built from such contracts Schär (2021)
- Surveys: technology and governance Böhme, Christin, Edelman & Moore (2015), Lin, Pele & Ren (2025); economics Härdle, Harvey & Reule (2020)

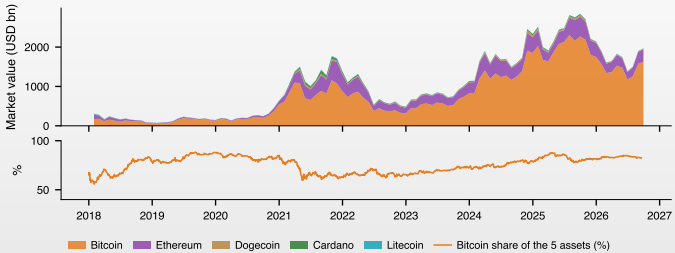
## Seven-Day Crypto Data Next to Five-Day Markets

| Series | Content | Period |
|---|---|---|
| Crypto prices | Bitcoin, Ethereum, Solana, Dogecoin, Cardano, Litecoin; daily, 7 days a week | 2018–2026 |
| Stablecoins | Tether (USDT), USD Coin (USDC), Dai (DAI) prices; supply by coin (DefiLlama) | 2019–2026 |
| Coin supply | Coins in circulation, daily (Coin Metrics Community Data) | 2018–2026 |
| DeFi | Total value locked, all chains and by chain (DefiLlama) | 2020–2026 |
| ETFs and shares | spot ETFs on Bitcoin (IBIT) and Ether (ETHA), Coinbase, Strategy; S&P 500, Nasdaq 100 (ETF QQQ), gold (XAU/USD, USD per ounce) | 2018–2026 |
| Asset universe | 26 assets: cryptos, equity indices, currencies, commodities, bonds | 2019–2026 |

- Daily closes up to 18 September 2026; crypto on its own 7-day calendar
  - analyses that combine assets use common days
  - prices aligned on common trading days first, then returns
  - some tests use longer histories and extra series


▸ Appendix: longer histories

# Five Coins, One Dominant Asset




- **Market value** of coin $i$ at month end $t$: $M_{i,t} = P_{i,t} \times Q_{i,t}$
  - $P_{i,t}$: the price in USD
  - $Q_{i,t}$: the coins in circulation
  - Bitcoin: 1.63 trillion USD on 18 Sep 2026
- The five coins: 1 970 billion USD today, peak 3 135 billion (6 Oct 2025)
  - the value moves with prices, supply changes slowly
- Bitcoin's share: 82.5% today, between 55.8% (1 Feb 2018) and 88.5% (5 Sep 2019)

MFM_ch16_market_value ▶ Interactive



▸ Appendix: market concentration

## Stylised Facts: the Question and the Choices

- **The question**: do crypto returns share the stylised facts of equities, on a different scale?
  - heavy tails, volatility clustering, deep drawdowns
- Why it matters: VaR, ES and every confidence interval depend on how heavy the tail is
  - VaR: value at risk; ES: expected shortfall
  - If the fourth moment is infinite, the sample kurtosis estimates no parameter: we need the tail index
- Comparison assets
  - the S&P 500, the reference risk asset
  - gold, the asset Bitcoin is compared with as "digital gold"
- Sample 2018–2026: one window for all assets
  - with the March 2020 crash, the 2022 crypto winter and the spot ETF launch
- Calendar: each asset's statistics on its own calendar
  - 7 days for crypto, 5 for shares
  - pairs of assets on common days

## The Measures: Return, Volatility, Skewness, Drawdown, Clustering

- Daily log return in %: $r_t = 100\,(\ln P_t - \ln P_{t-1})$
  - $P_t$: the close on day $t$
  - each series on its own calendar
- Annualised volatility $\sigma_a = s\sqrt{A}$
  - $s$: sample standard deviation of $r_t$; $A$: return periods per year, about 365 for crypto and 252 for shares
- Skewness $S = \overline{(r - \bar{r})^3}/s^3$, a bar denoting the sample mean: $S < 0$ means a longer left tail
- Maximum drawdown $\mathrm{MDD} = \min_t(P_t / \max_{u \le t} P_u - 1)$
  - the deepest fall from the running peak, between $-100\%$ and 0
- Volatility clustering $\hat{\rho}_1(|r|)$: sample autocorrelation of $|r_t|$ at lag 1
  - Chapter 1
  - $> 0$: large moves tend to follow large moves

## Historical VaR and ES

- Historical daily $\mathrm{VaR}_{1\%} = -q_{1\%}(r)$ and $\mathrm{ES}_{2.5\%} = -\frac{1}{0.025}\int_0^{0.025} q_u(r)\,du$
  - Chapter 7
  - $q_p(r)$: empirical $p$-quantile of the returns
  - historical ES: the integral of the empirical quantile function, with a fractional weight on the boundary observation
  - Why two levels: the Basel market-risk rules replaced VaR at 1% by ES at 2.5% (Basel Committee, MAR33, para. 33.3); under the Normal distribution both are close, $2.326\sigma$ and $2.338\sigma$; $\mathrm{ES}_p \geq \mathrm{VaR}_p$ holds at the same $p$, not across levels

## Estimating the Tail Index: the Hill Estimator

- Power-law tail of the daily loss $L = -r$: $P(L > x) \sim Cx^{-\alpha}$ for large $x$
  - $C > 0$: a scale constant
  - $\alpha$: the **tail index**; smaller $\alpha$ = heavier tail; the moment $E|r|^q$ is finite only for $q < \alpha$
- Hill estimator Hill (1975) from the $k$ largest losses: $\hat{\alpha}_H = \left(\frac{1}{k}\sum_{j=1}^{k} \ln(L_{(j)}/L_{(k+1)})\right)^{-1}$
  - $L_{(1)} \geq L_{(2)} \geq \ldots$: the losses sorted from the largest; $k = 5\%$ of the losses; $L_{(k+1)}$ acts as the threshold
  - Large ratios $L_{(j)}/L_{(k+1)}$ on average give a small $\hat{\alpha}_H$: a heavy tail
  - Standard error (SE) $\mathrm{SE} = \hat{\alpha}_H/\sqrt{k}$ for independent and identically distributed (i.i.d.) data; 95% confidence interval (CI) $\hat{\alpha}_H \pm 1.96\,\mathrm{SE}$ (conditions and dependence in the Appendix)

# Crypto vs Shares and Gold, 2018–2026 (1/2)

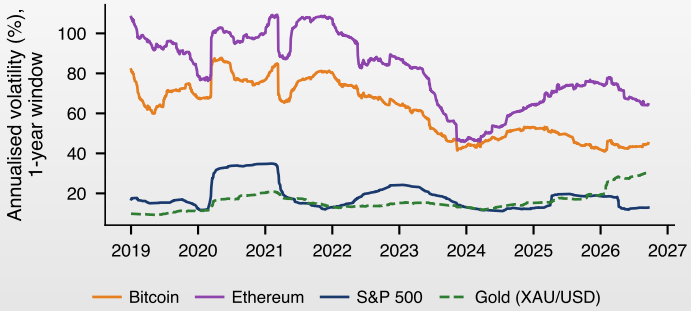
| | $n$ | $\sigma_a$ (%) | $S$ | MDD (%) | $\hat{\rho}_1(\lvert r\rvert)$ | $\hat{\alpha}_H$ | 95% CI | $\mathrm{VaR}_{1\%}$ | $\mathrm{ES}_{2.5\%}$ |
|---|---|---|---|---|---|---|---|---|---|
| Bitcoin | 3 182 | 64 | −0.95 | −82 | 0.17 | 2.73 | [2.12, 3.33] | 10.2 | 10.3 |
| Ethereum | 3 182 | 84 | −0.82 | −94 | 0.16 | 2.67 | [2.08, 3.26] | 13.5 | 13.7 |
| Solana | 2 351 | 117 | −0.22 | −96 | 0.28 | 2.85 | [2.12, 3.58] | 14.5 | 17.2 |
| Dogecoin | 3 182 | 124 | 4.88 | −92 | 0.37 | 2.83 | [2.22, 3.45] | 13.8 | 16.4 |
| S&P 500 | 2 189 | 19 | −0.65 | −34 | 0.36 | 2.87 | [2.07, 3.67] | 3.4 | 3.8 |
| Gold | 2 269 | 17 | −0.55 | −28 | 0.13 | 2.75 | [1.99, 3.51] | 3.0 | 3.2 |

- Daily log returns in %, Solana from April 2020; VaR and ES are losses of the daily log return
- Bitcoin: 3.3 times the S&P 500's volatility
  - daily VaR 1% 10.2% in log terms, a loss of 9.7% of the position
- Hill indices 2.67–2.87 for all six assets
  - crypto is more volatile, but the data give no strong evidence of different tail indices
  - tests in the Appendix

## Crypto vs Shares and Gold, 2018–2026 (2/2)

- All six i.i.d. 95% CIs lie below 3.7: $\alpha < 4$, hence $E|r|^4 = \infty$, is plausible but not proven
- If $\alpha < 4$, the sample excess kurtosis estimates no finite parameter
  - $K = \overline{(r - \bar{r})^4}/s^4 - 3$
  - report $\hat{\alpha}_H$ with its CI
- What the result supports: description

# Volatility Over Time



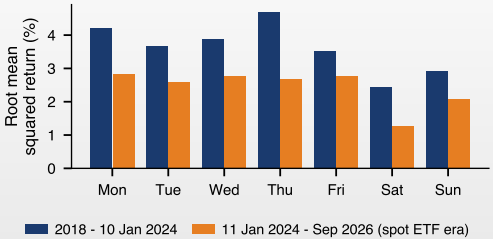

- Annualised standard deviation of daily log returns over a moving one-year window
- Bitcoin fell from 88% to 45% today, Ethereum from 109% to 64%
  - consistent with a maturing market, but the chart does not identify the cause
- Bitcoin is still 3.5 times as volatile as the S&P 500 (13%); gold reached 30% in 2026, a record for the sample

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▸ Appendix: drawdowns

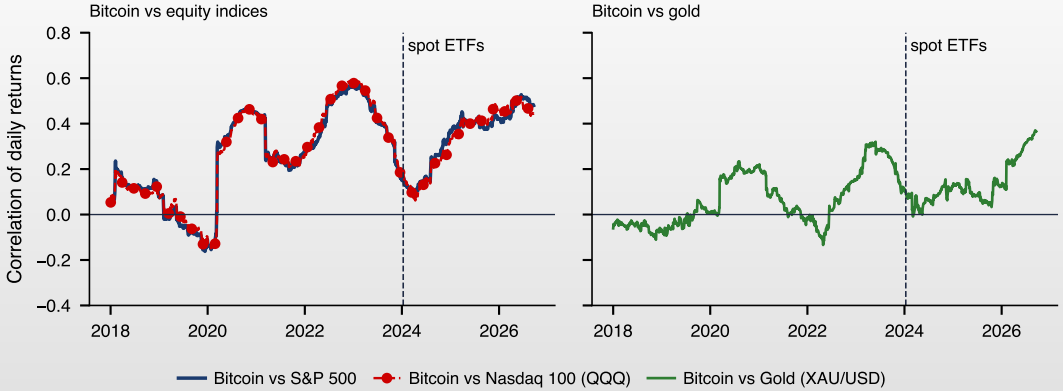
# A 7-Day Market with a Weekend Lull




- Root mean squared daily Bitcoin return by day of the week
  - weekend variance / weekday variance: 0.45 before and 0.39 after 11 January 2024
- Bitcoin trades on Saturdays, but macro news, US equity trading and ETF flows do not
  - all days calmer after the ETF launch (weekday root mean square 4.01% $\rightarrow$ 2.73%)
- Annualise with 365 days, but do not treat all days as identical
  - did the ratio change? Seminar 16, B1

MFM_ch16_weekday ▶ Interactive

# Is Bitcoin Moving with Shares? (1/2)




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## Is Bitcoin Moving with Shares? (2/2)

- Chart: correlation of daily log returns over a moving window of 250 common days
- Whole-period correlations, Bitcoin vs S&P 500
  - 0.02 (2017–2019, $n = 754$), 0.39 (2020–2023, $n = 1\,006$)
  - 0.38 since the ETFs ($n = 674$ days)
- Nasdaq 100 like the S&P 500 (0.38); gold $-0.02$, 0.12, 0.22
  - a smaller diversification benefit than before 2020 Bouri et al. (2017), Liu & Tsyvinski (2021)
- The chart suggests a break in early 2020, not in January 2024: tested next

MFM_ch16_correlation

## Bitcoin's Beta to the S&P 500: the CAPM

- CAPM (capital asset pricing model): $E[r_i] - r_f = \beta_i\,(E[r_m] - r_f)$, with $\beta_i = \mathrm{Cov}(r_i, r_m)/\mathrm{Var}(r_m)$
  - $r_f$: the risk-free rate; $r_m$: the market return (here the S&P 500); $r_i$: Bitcoin's return
- Why beta and not only the correlation: $\beta = \rho\,\sigma_{\mathrm{BTC}}/\sigma_{\mathrm{SPX}}$
  - The correlation says how tight the link is; beta says how much Bitcoin moves per 1% move of the S&P 500 (the hedge ratio)
  - On the common days of 2016–2026: $\rho = 0.26$ and $\sigma_{\mathrm{BTC}}/\sigma_{\mathrm{SPX}} = 3.7$ give $\beta = 0.95$
- Estimation: $\hat{b}$ from $r_t^{\mathrm{BTC}} = a + b\,r_t^{\mathrm{SPX}} + u_t$, daily log returns in %


▸ Appendix: HAC errors for beta

# Did the Bitcoin–Equity Link Break in 2020? (1/2): Model and sup-Wald

- The beta regression on 2 693 common days since January 2016
  - $(a, b)$ allowed to shift at an unknown date $\tau$: one pair before $\tau$, another after
- $W(\tau)$: the Wald statistic of $H_0$ (the same $(a, b)$ before and after $\tau$)
  - with Newey–West covariances (8 lags) in each regime
- sup-Wald Andrews (1993): $\sup_{\tau \in [0.15T, 0.85T]} W(\tau)$, the largest $W$ over the central 70% of the $T$ days
  - Under $H_0$ the sup is not $\chi^2_2$, because $\tau$ is chosen to make $W$ large
  - 5% critical value 11.55, simulated from the limit distribution (Appendix)
- Several breaks at unknown dates: the sequential tests and CIs of Bai & Perron (1998)
  - the test of a second break, $\sup F(2 \mid 1) = 10.33$, $p = 0.16$: no further break detected at 5%

# Did the Bitcoin–Equity Link Break in 2020? (2/2): the Result

- sup $W = 20.12$: above the 5% critical value 11.55
  - $p = 0.0008$, from 20 000 simulations of the limit distribution
  - so the no-break hypothesis is rejected
  - Newey–West covariances require finite moments of the score $x_t u_t$ and weak dependence, conditions the tails of Chapter 1 question
  - The maximum is reached on 14 Aug 2017, two trading days from the edge of the admissible range, far from the estimated break date
  - The rejection does not rest on that point: at the estimated date $W = 18.81$, above the 1% critical value 15.18
- Least-squares break date: 9 Mar 2020
  - the $\tau$ that minimises the sum of squared residuals (SSR) of the two regimes
  - 95% CI Bai (1997): 11 Feb 2019 – 29 Dec 2021, an interval that excludes January 2024
  - the interval is asymmetric: the S&P 500 variance rises from 0.75 to 1.61, the residual variance falls from 20.9 to 12.5
  - At the ETF launch $W = 2.31$
- Correlation: 0.01 before, 0.40 after
  - difference 0.39, 95% CI $[0.24, 0.46]$ (block bootstrap, at the estimated date)
  - adjusted for volatility (Forbes–Rigobon, in Further Topics), the 2020–2023 correlation falls from 0.39 to 0.23, but the rise stays positive
- What the result supports: description
  - the one-break model dates the change in the coefficients $(a, b)$; it does not identify a mechanism
  - $\rho = b\,\sigma_{\mathrm{SPX}}/\sigma_{\mathrm{BTC}}$ also depends on the variances: a break in $(a, b)$ is not a break in the correlation

## Beta Before and After the Break, with HAC Errors

- The same regression in each regime, Newey–West SE: 1 051 common days before 9 Mar 2020, 1 642 after
- Before: $\hat{b} = 0.07$, HAC SE 0.22 (OLS SE 0.16): $t = 0.3$, beta not significantly different from 0
- After: $\hat{b} = 1.22$, HAC SE 0.14 (OLS SE 0.07): $t = 8.6$
- HAC SE are 1.4 times (before) and 2.1 times (after) the OLS SE: OLS intervals would be far too narrow
  - HAC SE handle dependence, but require finite moments of the score $x_t u_t$, a condition the tails of Chapter 1 question
- Since the ETFs (674 days): $\hat{b} = 1.20$, HAC SE 0.20
  - in 2020–2023 $\hat{b} = 1.15$, HAC SE 0.17
  - the launch shows no change in beta

## Crypto Indices: the Question and the Choices

- **The question**: how do we build a benchmark for a market with many coins, dominated by one?
- Why it matters: the crypto market return is the return of such an index
  - in the GRS test (Section 7) and in the case study
- Market-value weights: the index is replicable
  - it holds the coins in the proportions the market holds them
- The divisor: keeps the level continuous when members or quantities change
- Number of constituents: chosen from the data by the CRIX rule, not fixed in advance
- Main threat: survivorship bias, if the universe of coins is chosen ex post

# A Crypto Index and Its Divisor



Humboldt-Universität zu Berlin, where CRIX was developed

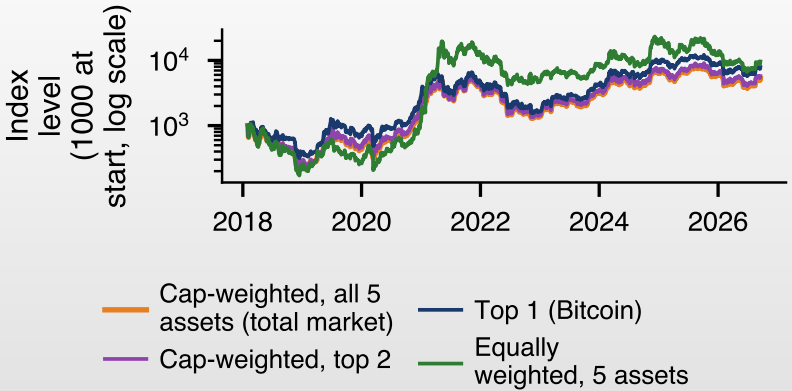
Photo: Diego Delso (2016); CC BY-SA 4.0; Wikimedia Commons

- **CRIX** Trimborn & Härdle (2018): a market-value weighted benchmark of the crypto market, published at thecrix.de
  - Review of crypto indices: Găman, Pele, Härdle & Będowska-Sójka (2026)
- Level on day $t$ of reallocation period $\tau$: $I_t = \sum_{i=1}^{k} P_{i,t}\, Q_{i,\tau} / D_\tau$
  - $k$: number of constituents; $P_{i,t}$: price of coin $i$ on day $t$; $Q_{i,\tau}$: its coins in circulation, fixed within the period; $D_\tau$: the **divisor**
  - Index return = weighted average of the constituents' returns, weights = value shares at the start of the period; a Laspeyres-type index (quantities fixed within the period)
- At a reallocation on day $t$ the divisor is reset so that the level does not jump
  - reallocation: new $Q$, new members
  - $D_{\tau+1} = \sum_i P_{i,t} Q_{i,\tau+1} / I_t$
  - Here: monthly reallocation, weights from the previous month end; worked examples: Seminar 16, A1, A4

## How Many Constituents? The CRIX Rule and the AIC

- Candidate index CRIX$(k, \beta)$, fitted to track the total market index TMI (all coins) Trimborn & Härdle (2018)
  - the $k_1$ largest coins enter with weight 1
  - the next $s = k - k_1$ with free weights $\beta = (\beta_1, \ldots, \beta_s)$, fitted to the data
- Tracking residual: $\hat{\varepsilon}_t(k, \beta) = \Delta \ln \mathrm{TMI}_t - \Delta \ln \mathrm{CRIX}_t(k, \beta)$, the daily gap between the two index log returns
- Akaike information criterion (AIC) $\mathrm{AIC}(k) = -2 \log L\{\hat{\varepsilon}(k, \hat{\beta})\} + 2s$
  - the rule picks the $k$ with the smallest AIC
  - $L$: likelihood of the residuals; $2s$: penalty for the free weights
  - $k$ reviewed quarterly on the last three months of data, members reallocated monthly by market value
  - Two stopping rules, ECRIX (first rise of the AIC) and EFCRIX (global minimum); likelihood details in the Appendix
- Here TMI holds only the five coins with public supply, so $k \leq 4$

## Crypto Indices Since 2018 (1/2)




MFM_ch16_crix ▶ Interactive

# Crypto Indices Since 2018 (2/2)

- Log scale; total-market index 21.4% a year over 8.6 years, volatility 65.7%
  - Bitcoin alone did better: the other coins lost weight and value
- Equal weights: 30.0% a year at 82.5% volatility
  - driven by 2021, when the equal-weighted index rose 998% and the value-weighted index 113%
  - a few extreme months, not a stable premium
- The five coins are chosen ex post, among today's survivors
  - the index returns are biased upwards (survivorship bias, in Further Topics; Będowska-Sójka, Wójcik & Pele (2026))
- What the result supports: description
  - an ex post comparison on five survivors, not evidence that one weighting rule is better

MFM_ch16_crix



Appendix: survivorship bias, weights

## Stablecoins: the Question and the Choices

- **The question**: how tightly do stablecoins hold their 1 USD peg, and what mechanism holds it?
- Why it matters: they are the cash leg of the crypto market
  - 303 billion USD on 18 Sep 2026
  - a peg can be lost over one weekend
- The coins: Tether, USD Coin and Dai, with daily prices over the whole sample
  - Tether and USD Coin: the largest fiat-backed stablecoins; Dai: crypto-backed
  - Sky Dollar, larger today, exists only since 2024
- Sample 2021–2026: earlier quotes are thin and noisy
- The model: a threshold AR, not a linear AR(1)
  - arbitrage creates a band around the peg
  - so the speed of reversion depends on the size of the deviation

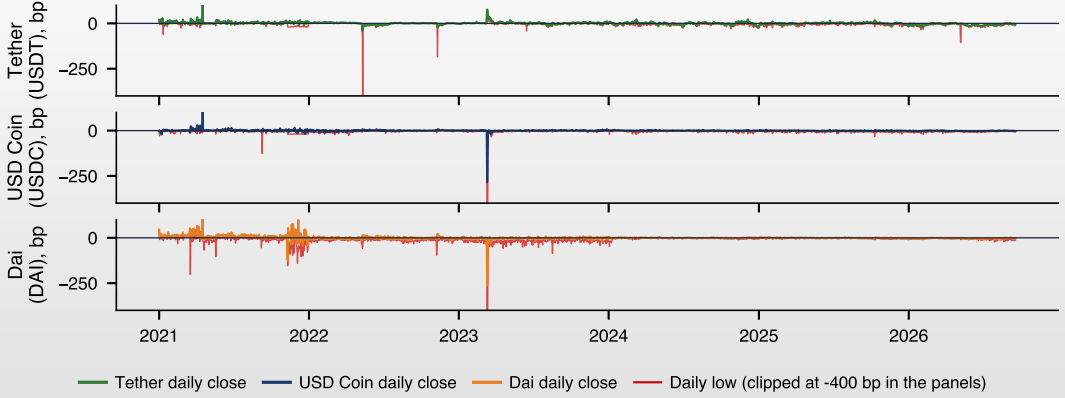
# How a Stablecoin Holds 1 USD

| Design | Backing | Examples (USD bn, 18 Sep 2026) |
|---|---|---|
| Fiat-backed | Bank deposits and short US Treasury bills held by the issuer (fiat: government-issued currency, here USD) | Tether (USDT) 183.33, USD Coin (USDC) 73.92, PayPal USD (PYUSD) 2.81 |
| Crypto-backed | Over-collateralised crypto deposits or hedged positions, managed by smart contracts | Sky Dollar (USDS) 6.49, Ethena USDe 4.74, Dai (DAI) 4.77 |
| Algorithmic | No external collateral: supply expands and contracts against a second token | TerraUSD (collapsed in May 2022) |

- Authorised traders **mint** and **redeem** with the issuer
  - mint: pay 1 USD, receive 1 coin; redeem: return 1 coin, receive 1 USD
  - arbitrage keeps the market price $P$ (USD) in the band $1 - c_r \leq P \leq 1 + c_m$
  - $c_r$, $c_m$: the redemption and minting costs per coin, in USD
  - Below the band: buy and redeem; above: mint and sell (evidence: Lyons & Viswanath-Natraj (2023); Tether issuance and Bitcoin prices: Griffin & Shams (2020))
- The band breaks when redemption stops or the backing is doubted
  - reserve and redemption rules: MiCA, Regulation (EU) 2023/1114, GENIUS Act, Public Law 119-27


▸ Appendix: stablecoin supply

# Deviations from 1 USD (1/2)




MFM_ch16_stablecoins ▶ Interactive

# Deviations from 1 USD (2/2)

- Daily close and daily low minus 1 USD, in basis points (1 bp = 0.01%), 2021–2026
  - Lows below −400 bp are cut at the panel edge
- Standard deviation of the close, 2021–2026: 7.3 bp (Tether), 7.5 bp (USD Coin), 10.8 bp (Dai)
  - Beyond 50 bp on less than 1% of the days
- Stress is rare but deep: USD Coin's low −1 226 bp (11 Mar 2023), Tether's −515 bp (12 May 2022)
- A stablecoin is a claim with a short put on the reserves
  - per coin the holder gets $\min(1, R) = 1 - \max(1 - R, 0)$
  - $R$: the reserve value per coin in USD
  - A sure 1 USD minus a put on $R$ with strike 1, sold to the issuer: nothing happens most days, a large loss is possible on a few

MFM_ch16_stablecoins

# Arbitrage Bands Imply a Threshold AR (1/2): Deviation and Model

- Deviation from the peg in basis points: $d_t = 10^4(P_t - 1)$
  - $P_t$: the daily close in USD
  - $P_t = 0.997$ gives $d_t = -30$ bp
- Threshold autoregression (TAR), the equilibrium form of Balke & Fomby (1997):

$$d_t = \begin{cases} \phi_{\text{in}}\, d_{t-1} + \varepsilon_t, & |d_{t-1}| \le c \\ \phi_{\text{out}}\, d_{t-1} + \varepsilon_t, & |d_{t-1}| > c \end{cases}$$

  - $c$: the half-width of the band in bp; $\varepsilon_t$: a shock with mean 0
  - Inside the band arbitrage does not pay (weak pull); outside it does (fast reversion)
  - $\phi_{\text{in}}$, $\phi_{\text{out}}$: persistence inside and outside the band
  - Half-life $\ln 0.5 / \ln|\phi|$: the number of days in which a deviation halves

# Arbitrage Bands Imply a Threshold AR (2/2): Threshold and Linearity Test

- $\hat{c}$ by least squares over a grid of $|d_{t-1}|$ values, keeping at least 15% of the days in each regime
  - $\hat{c}$ is an empirical regime boundary: it equals the cost, $10^4 c_r$ bp, only under a symmetric band ($c_r = c_m$) and no other frictions
- $H_0$: $\phi_{\text{in}} = \phi_{\text{out}}$: one linear AR(1), no band; $H_1$: the two persistences differ
  - For a given $c$: $W(c)$, the Wald statistic of $H_0$ with White standard errors White (1980) (robust to heteroskedasticity); the test uses $\sup_c W(c)$, the largest over the grid
- Under $H_0$ the threshold $c$ does not enter the model: it is not identified
  - so $\sup_c W(c)$ is not $\chi^2_1$ (the Davies problem)
  - $\chi^2_1$ critical values are too small for a maximum over many $c$: the test would reject too often

# Threshold AR Estimates for the Three Stablecoins

| | $\hat{c}$ (bp) | $\hat{\phi}_{\text{in}}$ (SE) | $\hat{\phi}_{\text{out}}$ (SE) | sup $W$ | $p$ bootstrap | $p$ ($\chi^2_1$) | $\hat{\phi}$ linear |
|---|---|---|---|---|---|---|---|
| Tether | 8.6 | 0.86 (0.02) | 0.69 (0.11) | 7.4 | 0.084 | 0.006 | 0.73 |
| USD Coin | 2.9 | 0.59 (0.04) | 0.28 (0.03) | 32.5 | 0.001 | $< 0.001$ | 0.29 |
| Dai | 6.0 | 0.72 (0.08) | 0.32 (0.07) | 13.8 | 0.005 | $< 0.001$ | 0.33 |

- USD Coin and Dai: linearity rejected in the AR(1), with independent multipliers
  - reversion is much faster outside the band (half-life 0.5 vs 1.3 days for USD Coin)
  - USD Coin, $p = 0.001$: the smallest value possible with $B = 999$ replications, not a bound on the true probability
  - Without 9–16 March 2023, USD Coin gives $p = 0.29$ (bootstrap) but 0.019 ($\chi^2_1$): one stress week carries the evidence, and $\chi^2$ over-rejects (step by step: Seminar 16, B5)
- Tether: the widest band ($\hat{c} = 8.6$ bp) and slow reversion inside it
  - but linearity is not rejected at 5%
  - consistent with fewer direct redeemers Ma, Zeng & Zhang (2025)
- What the result supports: description
  - $\hat{c}$ sits near the trimming limit (USD Coin, Tether): the half-lives depend on this imprecise estimate, the linearity test does not



▸ Appendix: the band width

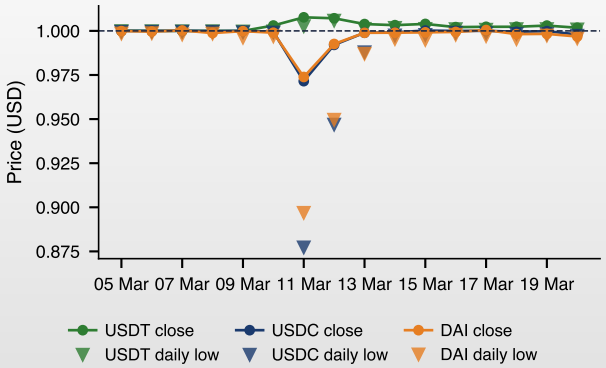
# Threshold AR: Inference with Serial Dependence

- The $p$-values of the table are conditional on the AR(1) and on independent multipliers $\eta_t$
  - lag-1 autocorrelation of the score $d_{t-\mathbf{1}}\hat{e}_t$: $-0.20$ (Tether), $-0.23$ (USD Coin), $-0.11$ (Dai)
  - the AR(1) residuals stay autocorrelated: Ljung–Box with 10 lags, $p < 0.01$ for all three
- Two corrections, with the threshold $\hat{c}$ re-estimated each time
  - multipliers held constant on blocks of 13 consecutive days (dependent wild bootstrap, Shao (2010))
  - lagged differences $\Delta d_{t-j}$ added until Ljung–Box no longer rejects at 5%

| | $p$ AR(1), independent | $p$ AR(1), blocks | $p$ with lags, blocks | lags | Holm |
|---|---|---|---|---|---|
| Tether | 0.084 | 0.083 | 0.223 | 5 | 0.446 |
| USD Coin | 0.001 | 0.001 | 0.001 | 2 | 0.003 |
| Dai | 0.005 | 0.011 | 0.376 | 8 | 0.446 |

- Holm: the family of the three coins, on the column with lags and blocks
- Only the rejection for USD Coin survives both corrections; for Dai it depends on the AR(1) specification

# March 2023: a Bank Failure Hits USD Coin



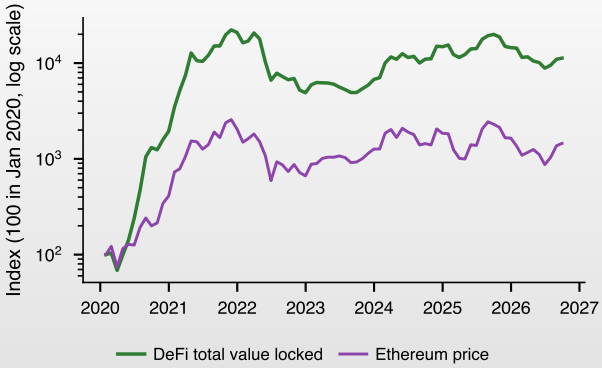

- Daily close (lines) and daily low (triangles), 5–20 March 2023
- Friday 10 March: Silicon Valley Bank (SVB) is closed
  - the FDIC (Federal Deposit Insurance Corporation) is appointed receiver FDIC (2023)
  - USD Coin held part of its reserves there Diop, Chevallier & Sanhaji (2024)
- Weekend: redemptions could not settle, the band disappeared
  - USD Coin's low 0.8774 USD on Saturday 11 March (close 0.9715)
  - Dai, crypto-backed, fell to 0.8970

MFM_ch16_stablecoins ▶ Interactive

## DeFi: the Question and the Choices

- **The question**: how does a liquidity pool set a price without an order book?
  - and who bears the cost of arbitrage?
- Why it matters: the pool is the counterparty to every swap
  - the liquidity provider holds a negative-gamma position
- The model: the constant product $x \cdot y = k$ (Uniswap v2)
  - the base form of weighted pools and concentrated liquidity
- The pool studied: Ether against a stablecoin, a single risky asset, so losses read directly in USD
- The benchmark: LVR (loss versus rebalancing), not simple holding
  - it separates the cost of arbitrage from the direction of the price
- Market size: total value locked (TVL)

# Total Value Locked in DeFi




- Total value locked (TVL), all chains (DefiLlama), and the Ether price
  - TVL: the value of tokens deposited in DeFi protocols
  - Month ends, 100 in January 2020, log scale
- TVL: 0.6 billion USD in January 2020, 177 billion at the peak (9 Nov 2021)
  - 88 billion today (50% below the peak)

MFM_ch16_defi ▶ Interactive



▸ Appendix: TVL by blockchain

# The Constant-Product Market Maker (1/2): the Pool and a Swap

- A pool holds reserves $x$ of token X and $y$ of token Y Adams, Zinsmeister & Robinson (2020)
  - e.g. X is Ether and Y is USD Coin
  - it accepts any trade that keeps $x \cdot y = k$
  - $k$: the invariant, set by the deposits; pool price (marginal): $p = y/x$ units of Y per X
- A trader sells $\Delta x$ of X and receives $\Delta y$ of Y
  - the pool keeps a fee $f$, so only $\gamma \Delta x$ enters the invariant, $\gamma = 1 - f$
  - $(x + \gamma \Delta x)(y - \Delta y) = xy \;\Rightarrow\; \Delta y = \dfrac{y \gamma \Delta x}{x + \gamma \Delta x}$
  - The fee $(1 - \gamma)\Delta x$ stays in the reserve, so the product of the reserves grows

## The Constant-Product Market Maker (2/2): the Execution Price

- Execution price $\Delta y/\Delta x = p\,\dfrac{\gamma}{1+\gamma\delta}$, with $\delta = \Delta x/x$ the trade size relative to the reserve
  - **Shortfall** against the pool price: $1-\gamma/(1+\gamma\delta)$: fee plus price impact; it grows with $f$ and with $\delta$
- Arbitrageurs keep $p$ in line with other exchanges: the pool price follows the market Angeris et al. (2019)


▸ Appendix: worked example, fee band

# Impermanent Loss and Loss Versus Rebalancing (1/3): the Positions

- A **liquidity provider** (LP) deposits $x_0$ of X and $y_0 = p_0 x_0$ of Y
  - half and half in value
  - receives a share of the pool
  - $p_0$: the price at the deposit; $p_1$: the market price later; $r = p_1/p_0$: the price relative (written $P_1/P_0$ in the chart)
  - Without fees, arbitrage moves the pool to $p_1$: from $xy = k$ and $y/x = p_1$, $x_1 = \sqrt{k/p_1}$, $y_1 = \sqrt{k\,p_1}$
- Two values in units of Y
  - the pool position $V_{\mathrm{LP}} = p_1 x_1 + y_1 = 2\sqrt{k\,p_1}$
  - simply holding the deposit $V_{\mathrm{H}} = p_1 x_0 + y_0$

## Impermanent Loss and Loss Versus Rebalancing (2/3): Impermanent Loss

- **Impermanent loss**, the relative shortfall of the pool against holding, using $k = p_0 x_0^2$:

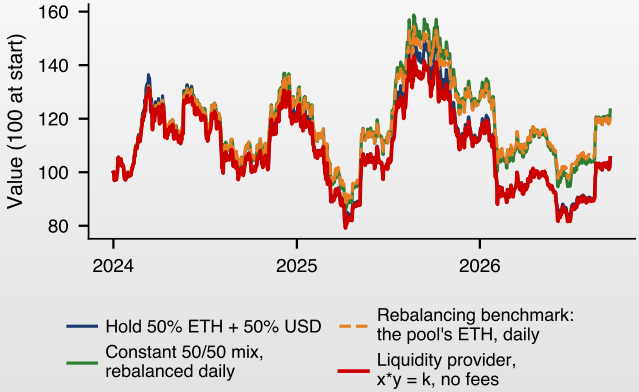
  $$\mathrm{IL}(r) = \frac{V_{\mathrm{LP}}}{V_{\mathrm{H}}} - 1 = \frac{2\sqrt{r}}{1+r} - 1 \leq 0$$

  - $\mathrm{IL} = 0$ only at $r = 1$; $r = 2$ or $r = 0.5$: $-5.72\%$; $r = 4$ or $0.25$: $-20.00\%$
  - Symmetric in $\ln r$: a doubling and a halving of the price cost the same
  - Small moves cost little: IL grows roughly with $(\ln r)^2$
- "Impermanent": it vanishes if the price returns to $p_0$
  - it becomes permanent if the LP withdraws before the price returns

# Impermanent Loss and Loss Versus Rebalancing (3/3): LVR

- A fairer benchmark than holding: a **rebalancing** portfolio Milionis et al. (2022)
  - it holds the pool's quantity of Ether at every instant and trades at market prices
  - The pool rebalances too, but at stale prices set by arbitrageurs: it always trades at a loss to them
- For $x \cdot y = k$, LVR grows at the instantaneous rate $d\mathrm{LVR}_t/(V_t\,dt) = \sigma_t^2/8$
  - loss versus rebalancing (LVR): the gap between the rebalancing portfolio and the pool
  - derivation by Itô in the Appendix
  - $V$: the pool value; $\sigma$: annualised volatility of the price; example: $\sigma = 60\%$ gives $0.36/8 = 4.5\%$ of the pool value a year
  - A weighted pool $x^w y^{1-w} = k$, with $w \in (0,1)$ the value weight of X ($w = 1/2$: constant product), loses at the rate $\sigma^2 w(1-w)/2$ (Seminar 16, A7)
- Fees must exceed $\sigma^2/8$ a year for a liquidity provider to beat this benchmark, in the frictionless model
  - not to earn a positive return
  - a volatility-selling business

# A Liquidity Provider in an Ether Pool (1/3)




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# A Liquidity Provider in an Ether Pool (2/3)

- Daily Ether prices from 1 January 2024
  - 100 invested half in Ether, half in USD
  - Pool with no fees; dashed: the LVR benchmark (the pool's Ether, rebalanced daily)
  - Constant 50/50 mix: weights reset daily (final value 123.3)
- Ether 2 352 $\rightarrow$ 2 611 USD ($r = 1.11$): impermanent loss only $-0.14\%$
  - Final values: holding 105.5, pool 105.4, LVR benchmark 122.4

# A Liquidity Provider in an Ether Pool (3/3)

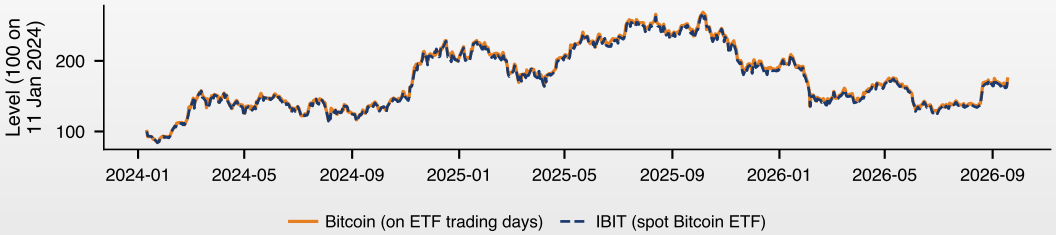
- Realised LVR 5.83% a year against $\sigma^2/8 = 5.79\%$ ($\sigma = 68.1\%$)
  - 95% block-bootstrap CI $[4.88, 6.98]\%$
  - the CI of a realised variance requires a finite fourth moment, which Chapter 1 questions
- Unlike impermanent loss, LVR accumulates with every move
  - in the frictionless model, fees would have to earn about 5.79% a year to match the benchmark
  - with fees, arbitrage happens only outside the band, and the loss is smaller Milionis, Moallemi & Roughgarden (2025)
- Why the LVR benchmark beats holding here
  - $R_T$: the LVR benchmark, $V_T^{\mathrm{LP}}$: the pool, $V_T^{\mathrm{H}}$: holding
  - $R_T - V_T^{\mathrm{H}} = \mathrm{LVR}_T + (V_T^{\mathrm{LP}} - V_T^{\mathrm{H}})$
  - here $122.4 - 105.5 \approx 17.0 + (105.4 - 105.5)$
  - The price ended near its start ($r = 1.11$), so the impermanent-loss term is tiny; after a strong trend ($r = 4$: IL $= -20.00\%$ of $V_T^{\mathrm{H}}$) it can outweigh LVR, and holding can come out ahead
- What the result supports: comparison of procedures
  - three positions on the same Ether price path, without fees

MFM_ch16_amm

## Spot ETFs: the Question and the Choices

- **The question**: does a spot ETF replicate the coin it holds?
  - and how much of a crypto equity's risk comes from Bitcoin?
- Why it matters: spot ETFs are how US stock-market investors hold Bitcoin and Ether
  - since January 2024
- The funds: IBIT and ETHA, the iShares (BlackRock) spot ETFs on Bitcoin and Ether BlackRock (2026)
  - same issuer and same 0.25% annual fee
- The measures: beta to the coin, tracking error and fee drift, daily and weekly
  - because the daily closes are not simultaneous
- The equities: two different channels of Bitcoin exposure
  - Coinbase: revenue from trading volume
  - Strategy: Bitcoin on the balance sheet
- Inference: Newey–West SE Newey & West (1987), because the residuals show volatility clustering

# IBIT Follows Bitcoin




- Bitcoin and IBIT on ETF trading days, both 100 on 11 Jan 2024
- A **spot ETF** holds the coins with a custodian and trades in US hours
- **Authorised participants** create or redeem shares against coins
  - this keeps the price within a cost band around the net asset value, as for a stablecoin
  - NAV: the fund's holdings at market prices, per share

MFM_ch16_etf ▶ Interactive

# Measuring How Well an ETF Tracks (1/2): Regression and Tracking Error

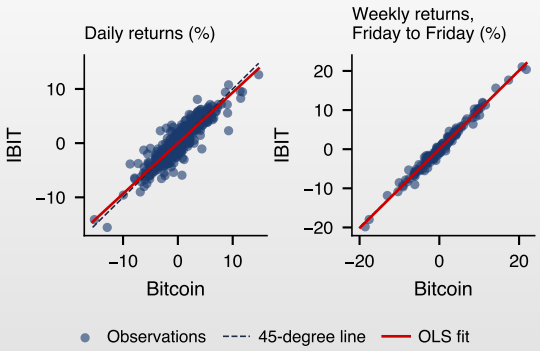
- Regression on the ETF's trading days: $r_t^{\mathrm{ETF}} = a + \beta\, r_t^{\mathrm{coin}} + u_t$
  - $r_t^{\mathrm{ETF}}$, $r_t^{\mathrm{coin}}$: log returns of the ETF and of its coin; a perfect tracker has $\beta = 1$, $R^2 = 1$
- **Tracking error** $\mathrm{TE} = \mathrm{sd}(r_t^{\mathrm{ETF}} - r_t^{\mathrm{coin}})\sqrt{A}$: annualised volatility of the return gap
  - sd: sample standard deviation; $A = 252$ daily, 52 weekly
- **Fee drift**: slope of $\ln(P_t^{\mathrm{ETF}}/P_t^{\mathrm{coin}})$ on time in years
  - the fund sells coins to pay its fee, so the ratio falls by about the fee each year

# Measuring How Well an ETF Tracks (2/2): Results

| | $\hat{\beta}$ daily | $R^2$ daily | $\hat{\beta}$ weekly | $R^2$ weekly | TE daily | TE weekly |
|---|---|---|---|---|---|---|
| IBIT / Bitcoin | 0.93 | 0.82 | 1.01 | 0.98 | 20.8 | 6.2 |
| ETHA / Ether | 0.94 | 0.83 | 0.99 | 0.98 | 29.9 | 9.0 |

- TE in % a year; weekly data show a near-perfect tracker
  - fee drift $-0.24\%$ a year (IBIT) and $-0.24\%$ (ETHA), close to the 0.25% fee
- Daily data understate the fit: the coin and the ETF close at different times
- IBIT trades 2.2 billion USD a day on average
- What the result supports: description

# Daily vs Weekly Returns and the Dimson Beta




- IBIT against Bitcoin: daily returns on common days (left), Friday-to-Friday returns (right); $R^2$ 0.82 vs 0.98
- The coin's daily close and the New York close are hours apart
  - Part of day $t$'s move reaches the ETF on $t \pm 1$: the daily OLS (ordinary least squares) beta is biased towards zero
  - Newey–West fixes the SE, not the bias

MFM_ch16_etf ▶ Interactive

# Crypto Equities: Coinbase and Strategy

- **Coinbase**: a US crypto exchange, revenue tied to trading volume
  - **Strategy** (formerly MicroStrategy): a software company holding Bitcoin as its main treasury asset
- Weekly two-factor regression, Newey–West SE, April 2021 – September 2026
  - $r_t^{\text{stock}} = a + \beta_{\text{BTC}}\, r_t^{\text{BTC}} + \beta_{\text{SPX}}\, r_t^{\text{S\&P}} + u_t$
  - $r_t^{\text{stock}}$, $r_t^{\text{BTC}}$, $r_t^{\text{S\&P}}$: weekly log returns of the stock, Bitcoin and the S&P 500
  - $\beta_{\text{BTC}}$, $\beta_{\text{SPX}}$: the stock's sensitivity to each, holding the other fixed
  - each SE requires finite moments of the score $x_t u_t$, a condition the tails of Chapter 1 question

| | $\hat{\beta}_{\text{BTC}}$ | SE | $\hat{\beta}_{\text{SPX}}$ | SE | $R^2$ | $\sigma_a$ (%) |
|---|---|---|---|---|---|---|
| Coinbase | 0.79 | 0.09 | 2.05 | 0.22 | 0.52 | 84 |
| Strategy | 1.18 | 0.11 | 1.02 | 0.24 | 0.63 | 89 |

- Strategy: 1.18 times Bitcoin plus 1.02 times the S&P 500
  - a leveraged Bitcoin position with extra equity risk
- Coinbase: $\hat{\beta}_{\text{SPX}} = 2.05$: a high-beta US stock, not a Bitcoin proxy
- What the result supports: association


▸ Appendix: rolling beta to Bitcoin

## The Asset Class: the Question and the Choices

- **The question**: are cryptos a separate asset class?
  - does a crypto market factor explain their mean returns?
- Why it matters: a separate class needs its own allocation and benchmark
  - an incomplete factor model leaves nonzero alphas
- Two views: pricing and distribution
  - pricing: a factor model tested by GRS and Fama–MacBeth
  - distribution: a map of return features, as in Pele et al. (2023)
- Test assets: eight large coins, weekly excess returns from February 2018, $T = 450$
  - Bitcoin, Ethereum, XRP, BNB, Cardano, Dogecoin, Litecoin, Chainlink
  - excess returns over the 4-week bill (DTB4WK)
- Market factor: the excess return of the index of Section 3
- Weekly frequency: avoids matching 7-day and 5-day calendars and follows the case study
- The map: one window before and one after the spot ETF launch
  - to see whether institutionalisation moved cryptos towards the other classes

## Testing a Factor Model (1/2): the Time-Series Regression

- Time-series regression for each test asset $i = 1, \ldots, N$: $r_{it}^e = \alpha_i + \beta_i' f_t + \varepsilon_{it}$, $t = 1, \ldots, T$
  - $r_{it}^e$: excess return of asset $i$ in week $t$ (over the Treasury bill); $f_t$: vector of the $K$ traded factor returns; $\beta_i$: the asset's loadings; $\varepsilon_{it}$: residual
  - $\alpha_i$: the mean excess return the factors leave unexplained; a correct model has $H_0 \colon \alpha_1 = \cdots = \alpha_N = 0$

# Testing a Factor Model (2/2): the GRS Test

- Gibbons, Ross & Shanken (1989): $\mathrm{GRS} = \dfrac{T-N-K}{N} \dfrac{\hat{\alpha}' \hat{\Sigma}^{-1} \hat{\alpha}}{1 + \bar{f}' \hat{\Omega}^{-1} \bar{f}} \sim F(N, T-N-K)$ under $H_0$
  - $\hat{\alpha}$: vector of the $N$ estimated alphas; $\hat{\Sigma}$: residual covariance; $\bar{f}$, $\hat{\Omega}$: mean and covariance of the factors
  - Numerator: the alphas, weighted by the precision of the residuals; denominator: one plus the squared Sharpe ratio of the factors
  - A large GRS (small $p$-value from the $F$ distribution) rejects zero alphas
- $F$ is exact only for i.i.d. errors with the Normal distribution
  - with crypto tails, add a wild bootstrap under $H_0$ (▸ Appendix)

## Worked Result: GRS and Fama–MacBeth on Eight Coins

- GRS $= 2.16$: zero alphas rejected at 5%
  - $p = 0.029$ from $F(8, 450 - 9)$, $p = 0.018$ from the wild bootstrap with independent multipliers (with serial dependence: next slide)
  - without normality, the test requires finite moments of $\varepsilon_t$ and $f_t\varepsilon_t$ and weak dependence, conditions Chapter 1 questions
  - Largest alphas: Dogecoin $+99\%$ a year ($t = 1.23$), BNB $+51\%$ ($t = 1.97$, the only one at the 5% threshold, and only just)
  - With eight tests, Holm's correction needs $p < 0.05/8 \approx 0.006$ for the smallest $p$-value, and BNB has $p \approx 0.05$: the rejection is joint, not driven by one coin
- Fama–MacBeth: $\hat{\lambda} = 163\%$ a year, SE 147 (FM) and 156 (Shanken)
  - the mean excess return of the factor was 39% ($t = 1.81$)
  - Betas range only from 0.90 to 1.40: the premium is estimated very imprecisely in this cross-section
- Survivorship may contribute to the rejection; GRS does not identify its cause
  - coins chosen because they are large today
- What the result supports: description

## GRS and Fama–MacBeth: Inference with Serial Dependence

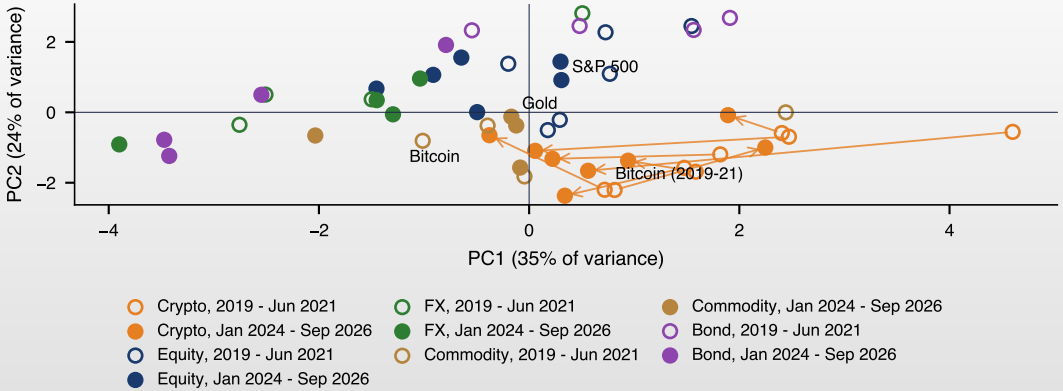
- The weekly residuals are not independent
  - Ljung–Box with 4 lags rejects no autocorrelation at 5% for 5 of the 8 coins
  - for the squared residuals: 6 of the 8 coins (volatility clustering)
- GRS with a dependent wild bootstrap (Shao (2010)): multipliers held constant on blocks of 4 weeks
  - $p = 0.038$, against 0.018 with independent multipliers: the rejection at 5% stands
- Fama–MacBeth, $\hat{\lambda} = 163\%$ a year
  - Newey–West SE of the weekly slopes (5 lags): 141
  - block bootstrap of 4 weeks, re-estimating both stages: SE 112, 95% CI $[-50, 381]\%$
  - the Shanken correction assumes i.i.d. errors and factors, which the diagnostics above do not support
- What the result supports: comparison of procedures
  - zero alphas are rejected at 5% under every calibration; the premium stays very imprecisely estimated

## A Separate Asset Class? The Evidence So Far

- Pele et al. (2023) compare cryptos with stocks, exchange rates, commodities, bonds and real-estate indices
  - Dimension reduction: most of the variation is explained by **tail**, **memory** and **moment** factors of log returns
  - Classification methods place cryptos in a separate class, mainly because of the tail factor
  - The tail factor (76% of the variance) loads on quantiles, conditional tail expectations, variance and the $\alpha$-stable parameters (Pele et al. (2023), Sect. 3.2): it mixes scale with shape, not a difference in Hill indices
  - Over time cryptos became more alike and more distinct from other assets
- Hedge or safe haven? Baur & Lucey (2010)
  - a hedge is uncorrelated or negatively correlated on average
  - a safe haven on the worst market days
- Early evidence: Bitcoin a diversifier, a hedge only in some periods Bouri et al. (2017)
  - tested with post-ETF data in Seminar 16, C2


▸ Appendix: replication method

# Cryptos Among Other Asset Classes (1/2)




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# Cryptos Among Other Asset Classes (2/2)

- First two principal components, PC1 and PC2 (35% and 24% of the variance)
  - Hollow: W1, filled: W2; FX in the legend: currencies
  - Arrows: each crypto
- PC1 measures volatility, kurtosis, drawdown depth and tail heaviness (loadings in the Appendix)
  - In W1 the cryptos sit apart on PC1, as in Pele et al. (2023); in W2 some of them overlap with the commodities
  - kurtosis exists only with a finite fourth moment, and the usual inference on it needs a finite eighth moment; Chapter 1 questions both
- Point estimates move towards the others
  - distance to equities 3.03 $\rightarrow$ 2.63, to commodities 2.26 $\rightarrow$ 1.86
  - Bitcoin's volatility 77% $\rightarrow$ 47%
- Cryptos became more alike (spread around their centre 2.13 $\rightarrow$ 1.43)
  - Median Hill index of the cryptos 2.32 $\rightarrow$ 3.26
- What the result supports: description
  - all 95% bootstrap CIs of the move contain 0 (Further Topics)

MFM_ch16_alt_assets


Appendix: significance of the move

## Case Study: Three Factors for Crypto Returns

- Liu, Tsyvinski & Wu (2022), *Journal of Finance*: the equity factor-model toolkit applied to crypto
  - portfolio sorts, factor regressions, GRS
  - on the whole cross-section of coins
  - Question: do a few crypto-specific factors price the expected returns of all coins, as market, size and momentum do for stocks?
- It builds on Liu & Tsyvinski (2021): aggregate crypto returns barely load on stock, currency or metal factors
  - but are predicted by crypto momentum and attention
  - The step: from the time series of three coins to the cross-section of 1,827 coins
- Why it matters: a factor model is the benchmark for every claimed crypto anomaly
  - and for judging crypto funds
  - if coins are segmented from equities, stock factors are the wrong benchmark
- Link to this chapter
  - the index of Section 3, the GRS test of Section 7
  - data-quality threats in the Appendix

# Case Study: Building Crypto Factors from 1,827 Coins

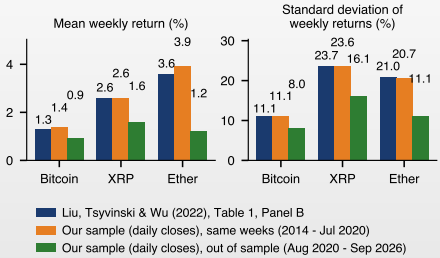
- Data: daily prices, volumes and market values of active and defunct coins (Liu, Tsyvinski & Wu (2022), Sect. 2)
  - from an aggregator of over 200 exchanges
  - market value of at least 1 million USD: 1,827 coins, first week of 2014 – July 2020
  - Weekly returns from daily closes: 52 weeks a year, the last one of eight days (nine in leap years); CMKT: value-weighted return of all coins minus the one-month bill rate
- Anomalies: 24 characteristics from the equity literature (Table 2)
  - they need only price, volume and market value
  - each week, value-weighted quintile portfolios and a top-minus-bottom spread
  - CSMB: bottom 30% minus top 30% of coins by market value; CMOM $= \frac{1}{2}(\mathrm{SH} + \mathrm{BH}) - \frac{1}{2}(\mathrm{SL} + \mathrm{BL})$ from $2 \times 3$ sorts on size and past three-week return (S/B: small/big coins; H/L: high/low past three-week return; Liu, Tsyvinski & Wu (2022), Sect. 4.1, Eq. 1)
- Tests: the ten significant spreads regressed on one-, two- and three-factor models (Tables 8–9)
  - joint GRS tests on all 24 spreads (Table 11)
  - The one-factor model, CMKT alone, is the crypto CAPM
  - Multiple testing: *p*-values adjusted for the many strategies tried (the error rate used: Appendix)

# Case Study: Size and Momentum Explain the Anomalies

| Result | Numbers in the paper | Reading |
|---|---|---|
| Weekly returns (Table 1) | mean, s.d.: market 1.3%, 11.2%; Bitcoin 1.3%, 11.1%; XRP 2.6%, 23.7%; Ether 3.6%, 21.0% | the market behaves like Bitcoin |
| Size (Table 3) | top minus bottom quintile $-5.8\%$ a week ($t = -2.45$) | premium in the smallest coins |
| Momentum (Table 4) | three-week winners minus losers 3.1% a week ($t = 2.65$) | winners keep winning |
| Crypto CAPM (Table 8) | all ten alphas significant; momentum $\alpha = 3.0\%$ ($t = 2.55$), $R^2 = 0.001$ | the market alone fails |
| Three factors (Table 9) | no alpha significant; momentum $\alpha = 1.1\%$ ($t = 1.41$), CMOM loading 1.12, $R^2 = 0.58$ | size and momentum explain the ten anomalies |
| Joint test (Table 11) | GRS on 24 spreads: $p = 0.033$ (Fama–French six factors), $p = 0.194$ (crypto three) | stock factors are the wrong benchmark |

- Momentum is a large-coin effect, the opposite of stocks
  - 3.2% a week above the median size, insignificant below (Table IA.10)
  - IA: the paper's Internet Appendix
- What the result supports: comparison of procedures

# Case Study: Table 1 Rebuilt on the Course Data




- Largest gap, the Ether mean: 0.26 i.i.d. standard errors (SE 1.29 points)
- Out of sample (319 weeks): means 0.9, 1.6, 1.2%
  - S.d. 8.0, 16.1, 11.1%: Ether's volatility roughly halves

- Weekly simple returns on the paper's calendar
  - Blue: Table 1, Panel B of Liu, Tsyvinski & Wu (2022)
  - Orange: our sample (daily closes), same weeks
  - Green: August 2020 – 16 Sep 2026

MFM_ch16_ltw ▶ Interactive

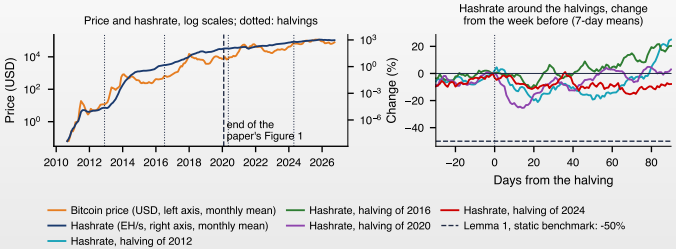
## Limitations and Out-of-Sample Tests

- **Limits** stated by the authors (Section 6.2)
  - A short sample, 2014 – July 2020, while investors were still learning about the technology: only the split-half check (same signs)
  - Most coins cannot be shorted; for the top 20 coins, after trading costs, about 60% of the returns of long–short strategies that short Bitcoin remain (Table IA.16)
  - With prices from three reputable USD exchanges, the market-value size spread is no longer significant (Table IA.15)
- **Open extension**, out of sample, August 2020 – September 2026 (the 2022 collapses, the spot ETFs)
  - Does three-week momentum among large coins survive?
  - Is CMKT still segmented from equities?
  - Crypto-CAPM alphas; the segmentation regression of footnote 16 of Liu, Tsyvinski & Wu (2022) on the global five factors Fama & French (2015) and five exchange rates; a break test at an unknown date (Seminar 16, C3)

## Case Study: The Bitcoin Price and Network Security

- Pagnotta (2022), *Review of Financial Studies*: an equilibrium model in which the Bitcoin price and the security of the network are determined jointly
  - users value coins taking into account the risk of an attack that would block transfers
  - miners invest computing power for the block reward, paid in bitcoin
- **Hashrate**: the total computing power of the miners, in hashes per second (1 EH/s $= 10^{18}$ hashes/s)
  - an attack succeeds only with computing power comparable to that of the honest miners
- The question: does Bitcoin's security depend on its price, not only on its technology?
- The paper's claims
  - the same technology admits multiple equilibria, with very different prices and security levels
  - the price–security feedback amplifies the volatility impact of fundamental shocks
  - it can produce boom–bust cycles not driven by fundamentals
  - the empirical implication: price and hashrate move together (Figure 1, Aug 2010 – Jan 2020)
- **Security budget**: the block reward plus the fees paid to miners
  - daily share of the block reward, Jul 2010 – Jan 2020: median 99.21%, mean 97.57% (the paper, Section 1.1, footnote 16, in the manuscript SSRN 3264448)
  - the same share in the Coin Metrics Community Data data: median 99.20%, mean 97.58%
  - every halving cuts the block reward in half

# Case Study: Bitcoin Price and Hashrate, 2010–2026




- Left: monthly means of the price (course data) and of the hashrate (Coin Metrics Community Data), Aug 2010 – Sep 2026
  - hashrate: 952 EH/s, mean of the 7 days to 18 Sep 2026
- Right: change of the hashrate from the week before each of the four halvings
  - dashed line: −50%, the static benchmark of Lemma 1 with the expected price, the cost and the number of miners unchanged
  - largest fall in the first 90 days: −25% (2020, day 14)

MFM_ch16_pagnotta ▶ Interactive

# Case Study: Lemma 1 and the Replication on the Course Data

- Lemma 1 of Pagnotta (2022): $m$ identical miners; the lemma allows a convex cost, here the case of a linear cost $c$ per unit of hashrate
  - equilibrium hashrate: $H^* = \frac{m-1}{m} \frac{\delta R \, \mathbb{E}_t[p_{t+\mathbf{1}}]}{c}$
  - $R$: the reward per block, in bitcoin; $p_{t+\mathbf{1}}$: the price at which the miner sells the reward; $\delta$: the discount factor
  - Prediction 1: the elasticity of $H^*$ to $R$ is 1, with $\mathbb{E}_t[p_{t+\mathbf{1}}]$, $c$ and $m$ held constant
  - Prediction 2: with the expected price, the cost and the number of miners unchanged, a halving halves $H^*$
- Data: the Bitcoin price from the course data; hashrate, coins issued, fees and number of blocks from Coin Metrics Community Data
  - USD reward per block on day $t$: $B_t = p_t \times$ (subsidy + fees) / number of blocks
  - the miners' daily revenue also moves with the number of blocks found on day $t$, which also enters the hashrate estimate
  - the current price $p_t$ stands in for the expected price $\mathbb{E}_t[p_{t+\mathbf{1}}]$
- The paper shows the series only (Figure 1); the replication adds three tests
  - regression of $\ln H$ on $\ln B$ and a linear trend, on monthly means: the trend absorbs the fall of the cost $c$ as hardware becomes more efficient
  - the Engle & Granger (1987) test: the null hypothesis is no cointegration between $\ln H$ and $\ln B$
  - the four halvings: the subsidy per block halves at blocks 210,000, 420,000, 630,000 and 840,000; the dates are the UTC days of these blocks
- Windows: the paper's (Aug 2010 – Jan 2020) and out of sample (Feb 2020 – Aug 2026)

# Case Study: At Halvings, Hashrate Falls Less than the Static 50% Benchmark

| Test | Course data | Interpretation |
|---|---|---|
| Levels, Aug 2010 – Jan 2020 | elasticity to $B$: 1.19 (HAC SE 0.21) with trend; 2.44 without trend ($R^2 = 0.91$) | association close to 1 |
| Other revenues, with trend | subsidy per block 1.21; daily revenue 1.15 | the same association |
| Levels, out of sample | elasticity: 0.03 (HAC SE 0.09) with trend; 0.82 without trend ($R^2 = 0.24$) | not significant next to the trend, which does not show that the link has gone |
| Engle–Granger | p-value: 0.79 (paper's window), 0.72 (out of sample) | no evidence of cointegration: the level regressions are descriptive |
| Halving, day 30: USD reward per block | 2012: −45%; 2016: −56%; 2020: −48%; 2024: −52% | close to −50% |
| Halving, day 30: hashrate | 2012: −10%; 2016: +4%; 2020: −5%; 2024: −8% | changes smaller than the static −50% benchmark |

- What the result supports: description
  - descriptive associations: they do not identify the effect of security on the price and do not test the multiple equilibria
- Lemma 1 is a static equilibrium condition: hardware already bought is a sunk cost, and miners stay active while the reward covers the cost of energy
- Limits: the hashrate is estimated from block times; miners anticipate the halvings; the expected price is not observed

## An Open Question and the Discovery Loop (1/2)

- Did the US spot Bitcoin ETFs (since 11 January 2024) integrate Bitcoin with equities?
  - In the mean?
  - In the tails?
  - In price discovery?
- A live question
  - Lee & Lee (2025) study the shifted links after the approval
  - Kang & Lee (2026) title their answer "institutionalization without integration"
- Formal version: $H_0 : \rho_{\text{post}} = \rho_{\text{pre}}$ against $H_1 : \rho_{\text{post}} > \rho_{\text{pre}}$
  - $\rho_{\text{pre}}$, $\rho_{\text{post}}$: the correlations before and after the launch
  - the same for the lower-tail dependence $\lambda_L$ of Chapter 6
  - lower-tail dependence: the probability of an extreme Bitcoin loss given an extreme equity loss
  - Falsified if the data choose a break far from January 2024, or if placebo dates give changes as large

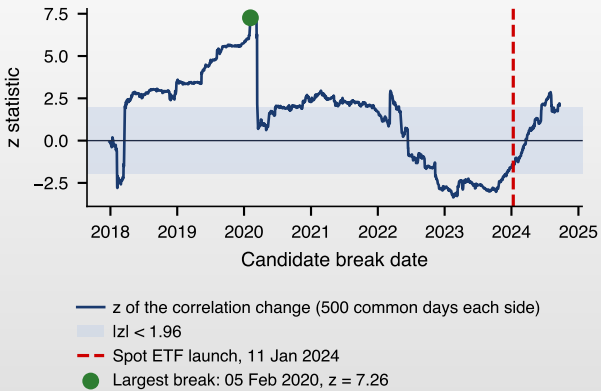
## An Open Question and the Discovery Loop (2/2)

- Identification: the launch is an event, not an experiment
  - it overlaps with the rate cycle, the April 2024 halving and flows that respond to prices
- The discovery loop in one pass, the assistant drafts and the researcher verifies
  - literature (resolve every DOI, the digital object identifier of a paper), competing hypotheses, code that first replicates $\hat{\rho} = 0.39$ and 0.38, a pre-registered robustness plan, a referee-style critique; reusable prompts in the Appendix

## What the Researcher Must Check

- **Calendars**: align prices on common trading days first, then compute returns
  - returns computed on each calendar and then matched pair Bitcoin's Sunday-to-Monday move with the S&P 500's Friday-to-Monday move
- **Closing times and volatility** Forbes & Rigobon (2002)
  - non-synchronous closes bias daily correlations towards zero, high volatility raises them
  - check weekly returns, the lead–lag correlation and the adjusted $\rho^*$
- **Survivorship and supply**: use the universe as it was on each date and the circulating supply, as in the CRIX rule
- **Invented references and numbers**
  - every DOI resolves on Crossref with the same title
  - every number is regenerated by code in the repository

# Mini-Case: Did the ETFs Break the Correlation? (1/2)




- Assistant's claim: "the spot ETFs integrated Bitcoin with equities"
  - $\hat{\rho} = 0.39$ in 2020–2023 ($n = 1\,006$), 0.38 since 11 January 2024 ($n = 674$)
- Difference $-0.01$, 95% CI $[-0.12, 0.13]$ (circular block bootstrap, blocks of 20 days)
  - the CI of a correlation requires a finite fourth moment, which Chapter 1 questions

ai_discovery_case.py ▶ Interactive

# Mini-Case: Did the ETFs Break the Correlation? (2/2)

- Chart: $z(\tau) = [\operatorname{atanh} \hat{\rho}_{\text{after}} - \operatorname{atanh} \hat{\rho}_{\text{before}}] / \sqrt{2/(h-3)}$
  - for each candidate date $\tau$
  - correlations over $h = 500$ days on each side of $\tau$
  - atanh (Fisher) makes each estimate roughly Normal with variance $1/(h-3)$, so $z \approx N(0,1)$ without a change
- Largest $z(\tau)$ on 5 Feb 2020 ($z = 7.26$); at the ETF date $z = -1.48$
  - 82% of candidate dates give a larger $|z|$
  - The scan points to early 2020, where the sup-Wald test places the break
- The $\pm 1.96$ band of the chart is pointwise
  - the maximum over more than a thousand overlapping dates needs its own critical value, as for sup $W$
- With heavy tails and volatility clustering the variance of $\operatorname{atanh} \hat{\rho}$ is typically larger than $1/(h-3)$
- So the scan is descriptive; the inference is the sup-Wald test
- What the result supports: description
  - the ETF launch is an event, not an experiment

## Your Turn: a Project Seed

- **Question**: did institutionalisation change Bitcoin's **tail** dependence and **price discovery** with US equities?
- **Replicate first**: $\hat{\rho} = 0.39$ and 0.38 from this chapter
  - then one headline estimate of Lee & Lee (2025) or Kang & Lee (2026) on their own sample
  - Extend: lower-tail dependence from copulas (Chapter 6), weekly returns, IBIT–Bitcoin lead–lag in US trading hours
- **Deliverables**: a pre-registered plan committed before the results
  - a notebook that regenerates every number
  - effect sizes with CIs and the full list of specifications
- **AI log** (templates)
  - prompts that shaped the design in `AI_USE.md`
  - each wrong claim (an invented DOI, returns on mismatched calendars) and the test that caught it in `AI_ERRORS.md`

## Key Takeaways

- Crypto returns: 3–7 times the volatility of shares, drawdowns above 80%
  - calmer weekends, heavy tails close to those of shares and gold
  - the one-break model dates the change in the coefficients to March 2020
  - the chosen comparison detects no rise in the correlation after the ETFs
  - beta to the S&P 500 rose from near 0 to above 1
- Indices: a Laspeyres-type index with a divisor
  - CRIX picks its size by an AIC on tracking residuals
  - value weights make crypto indices mostly Bitcoin
- Stablecoins: the peg is an arbitrage band that holds only while redemption works
  - a threshold AR fits it, tested with a bootstrap, not with $\chi^2$
- DeFi: $xy = k$ prices a swap as fee plus price impact
  - providing liquidity is selling volatility, at $\sigma^2/8$ a year
- ETFs track one for one once closing times are handled
  - cryptos remain a separate class in distribution, a risk asset in co-movement

# Check Your Understanding, and What Comes Next

## Retrieval questions

- Why is Bitcoin's sample kurtosis not reported?
- What replaces it?
- Why can a threshold or a break date not be tested with $\chi^2$ critical values?
- When does a stablecoin's arbitrage band stop holding the peg?
- Why does a liquidity provider fall behind the rebalancing benchmark even if the price returns to $p_0$?
- Why does a daily regression understate how well IBIT tracks Bitcoin?

## Next chapters

- Chapter 17, bubbles and crashes: explosive-root tests and log-periodic power laws, crypto booms as a main case
- Chapter 18, systemic risk: spillovers between crypto, banks and equity markets
- Revisit
  - VaR and ES for crypto (Chapter 7, Pele & Mazurencu-Marinescu-Pele (2019b))
  - Arbitrage, segmentation and liquidity across crypto exchanges (Chapter 10, Makarov & Schoar (2020), Agakishiev, Härdle & Pele (2025))
- Reading
  - Härdle, Harvey & Reule (2020), Franke, Härdle & Hafner (2019), Schär (2021)

## The Practical Decision

- **What I choose**
  - the Hill tail index with its CI instead of the sample kurtosis: all six 95% CIs lie below 3.7
  - beta and correlation with equities from the regime after the 9 Mar 2020 break ($\hat{b} = 1.22$), on common days
  - weekly returns to judge ETF tracking ($\hat{\beta}$ of 1.01 and 0.99)
  - for a stablecoin peg, a threshold AR tested by a bootstrap that keeps the dependence, not with $\chi^2$
- **Under which conditions**
  - redemption at the issuer works: otherwise the arbitrage band disappears (USD Coin, 11 Mar 2023)
  - for a liquidity provider: fees above $\sigma^2/8$, i.e. 5.79% a year in the Ether pool, the frictionless benchmark
- **What result would make me drop it**
  - a new sup-Wald test that dates a later break, after which beta returns towards 0
  - dependence-robust Hill CIs lying entirely above 4, for both tails: only then would the kurtosis have a finite parameter to estimate
  - on weekly data, an ETF beta far from 1 or a fee drift well above the 0.25% fee

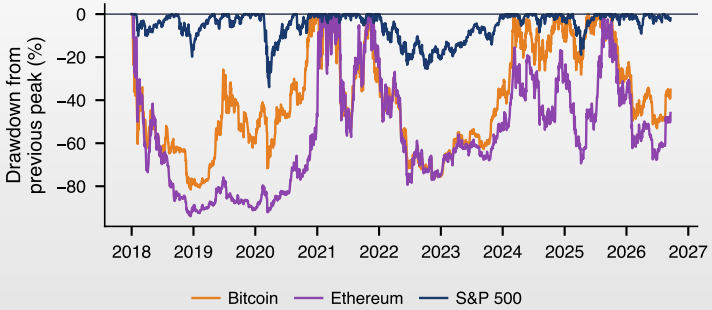
## Longer Histories and Extra Series

| Series | Period | Used for |
|---|---|---|
| Bitcoin and S&P 500, daily | from January 2016 | 2017–2019 correlations, the Forbes–Rigobon base period, the break test (2 693 common days) |
| Bitcoin and XRP; Ether | from January 2014; from 6 August 2015 (week 32) | the case study: Table 1 of Liu, Tsyvinski & Wu (2022) rebuilt, 52-week volatility |
| XRP, BNB, Chainlink, daily | weekly returns from 9 Feb 2018 | the GRS test on eight coins (Section 7), the case-study replication |
| 4-week Treasury bill, DTB4WK (FRED) | the same weeks as the GRS test | excess returns in the GRS test |
| One-month Treasury bill rate (Kenneth French Data Library) | the case-study weeks | the case study, definition of CMKT |
| Spot ETFs IBIT and ETHA | from 11 Jan 2024 and 23 Jul 2024 | ETF tracking and the Dimson beta |
| Coinbase and Strategy shares | weekly, April 2021 – September 2026 | the two-factor regressions of crypto equities |

## Five Coins, One Dominant Asset: Concentration

- Concentration: **Herfindahl–Hirschman index** $\mathrm{HHI} = \sum_i w_i^2$, $w_i = M_{i,t} / \sum_j M_{j,t}$ the value share of coin $i$
  - From 1/5 (equal shares) to 1 (one coin)
  - Today 0.71, an effective number of coins $1/\mathrm{HHI} = 1.42$
- The five coins: Bitcoin, Ethereum, Dogecoin, Cardano and Litecoin
  - Solana has no public circulating-supply series, hence no market value (Section 3)
- XRP and Chainlink are left out: their reported supply includes issuer-held tokens
  - XRP: the native token of the XRP Ledger
  - free float matters

# Drawdowns: How Deep and How Long




- Fall from the previous maximum, in %: Bitcoin was more than 50% below its peak on 36% of the days since 2018
  - Ethereum on 61%; the S&P 500 at most 34% (March 2020)
- Today: Bitcoin 35%, Ethereum 46% below their peaks
  - long "crypto winters" (2018–2020, 2022–2023) follow each boom (bubbles: Chapter 17)

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## Bitcoin's Beta to the S&P 500: HAC Standard Errors

- Newey–West SE Newey & West (1987): $\widehat{\mathrm{Var}}(\hat{a}, \hat{b}) = (X'X)^{-1}\,\hat{S}\,(X'X)^{-1}$
  - $\hat{S} = \hat{\Gamma}_0 + \sum_{l=1}^{L}(1 - \frac{l}{L+1})(\hat{\Gamma}_l + \hat{\Gamma}_l')$, with $\hat{\Gamma}_l = \sum_t \hat{u}_t \hat{u}_{t-l}\, x_t x_{t-l}'$ and $x_t = (1, r_t^{\mathrm{SPX}})'$, row $t$ of the matrix $X$
  - The Bartlett weights $1 - l/(L+1)$ keep $\hat{S}$ positive semi-definite
  - $L = \lfloor 4(T/100)^{2/9} \rfloor$ lags: 8 for $T = 2\,693$ days
- Why HAC (heteroskedasticity and autocorrelation consistent)?
  - with volatility clustering the variance of $u_t$ rises on turbulent market days
  - OLS SE assume a constant variance and uncorrelated errors
- Whole sample: $\hat{b} = 0.95$, HAC SE 0.13, against an OLS SE of 0.07
  - One beta for 2016–2026 assumes a stable link: first test for a break at an unknown date

# Correlation or Volatility? The Forbes–Rigobon Adjustment (1/2)

- Pearson correlation rises mechanically when the volatility of the driving market rises Forbes & Rigobon (2002)
  - with the same underlying link
- Adjusted correlation: the correlation the volatile period would show at the volatility of the calm period
  - $\rho^* = \dfrac{\rho}{\sqrt{1+\delta(1-\rho^2)}}, \qquad \delta = \dfrac{\sigma^2_{\mathrm{SPX,high}}}{\sigma^2_{\mathrm{SPX,low}}} - 1$
  - $\rho$: correlation of Bitcoin and S&P 500 returns in the volatile period; $\rho^*$: the adjusted correlation
  - $\sigma^2_{\mathrm{SPX,high}}$, $\sigma^2_{\mathrm{SPX,low}}$: variance of S&P 500 returns in the volatile period and in the calm base period (2017–2019)
  - $\delta$: relative rise of that variance; $\delta = 0$ gives $\rho^* = \rho$, $\delta > 0$ gives $\rho^* < \rho$
- Caveat: the adjustment assumes no endogeneity and no omitted common shocks
  - a pandemic hardly satisfies these assumptions

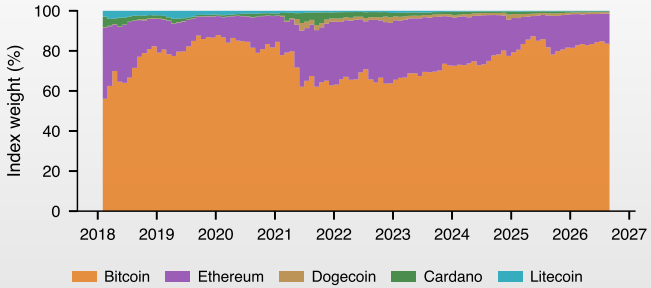
## Correlation or Volatility? The Forbes–Rigobon Adjustment (2/2)

- 2020–2023: S&P 500 volatility 23.1% vs 12.8% ($\delta = 2.24$); $\hat{\rho} = 0.39$ becomes $\rho^* = 0.23$
  - Since the ETFs: $\hat{\rho} = 0.38$, $\rho^* = 0.33$ ($\delta = 0.42$)
- Change against 2017–2019 ($\hat{\rho} = 0.02$): $\rho^*_{\text{2020–23}} - \hat{\rho}_{\text{2017–19}} = 0.21$
  - 95% block-bootstrap CI $[0.09, 0.33]$
  - The CI excludes 0: the adjusted correlation did rise
  - Under the adjustment's assumptions, 44% of the raw rise is attributed to volatility; the adjusted increase remains positive: stronger co-movement, not an identified mechanism

## Survivorship Bias: Indices of Survivors

- The universe at the start: coins that will survive ($S_i = 1$) and coins that will die ($S_i = 0$)
  - $\pi = P(S_i = 1)$
- Population mean: $E[\bar{r}_i] = \pi\, E[\bar{r}_i \mid S_i = 1] + (1 - \pi)\, E[\bar{r}_i \mid S_i = 0]$
  - $\bar{r}_i$: the mean return of coin $i$
- Bias of a survivors-only sample: $E[\bar{r}_i \mid S_i = 1] - E[\bar{r}_i] = (1 - \pi)\big(E[\bar{r}_i \mid S_i = 1] - E[\bar{r}_i \mid S_i = 0]\big)$
  - Positive when survival depends on returns: coins that crash are delisted Brown, Goetzmann, Ibbotson & Ross (1992)
  - Grows with the share of dead coins, $1 - \pi$, large in crypto
- Our indices use five coins chosen today
  - their returns are probably higher than those of a strategy one could have run in 2018
- Remedy: the point-in-time universe, dead coins included, as in Liu, Tsyvinski & Wu (2022)

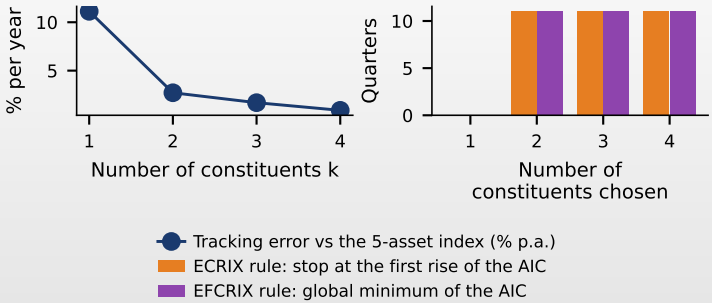
# Index Weights Over Time




- Value weights at each monthly reallocation, %: Bitcoin 75.7% on average, never below 56.1%, today 82.9%
  - Ethereum today 15.8%; the other three together today below 2%
- 82.9% against the 82.5% market-value share: index weights are fixed at the last reallocation
- Altcoins (coins other than Bitcoin) gain weight late in booms (2021) and lose it in busts
  - a value-weighted crypto index is mostly Bitcoin

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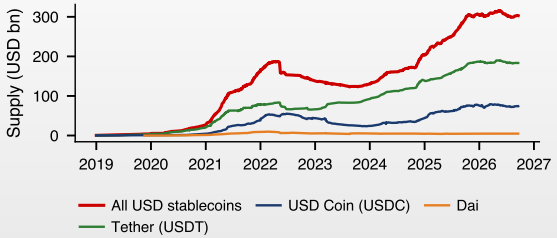
# Tracking Error and the CRIX Rule




- Left: tracking error of the top-$k$ index (weights 1) against the five-coin index
  - 11.1% a year for Bitcoin alone, 2.7% for two coins, 1.7% for three, 0.9% for four
- Right: the rule picks 2 coins in 11, 3 in 11 and 4 in 11 of 33 quarters (2 in the last)
  - ECRIX and EFCRIX agree in every quarter

MFM_ch16_crix

# The Rise of Stablecoins




- Supply in circulation (DefiLlama): from 4.2 billion USD in January 2020 to 303 billion on 18 Sep 2026
  - about 73 times: the cash leg of crypto trading and DeFi
- Tether 183 billion (61%), USD Coin 74 billion (24%)
  - for the 42 coins above 100 million USD at extraction (21 Oct 2026), valued on 18 Sep 2026, a universe chosen retrospectively: fiat-backed 278, crypto-backed 22.2, algorithmic 0.15 billion
- 2022–2023 dip: 187 billion (2 Apr 2022) to 123 billion (19 Aug 2023)
  - TerraUSD vanished, USD Coin fell from 55 to 23 billion

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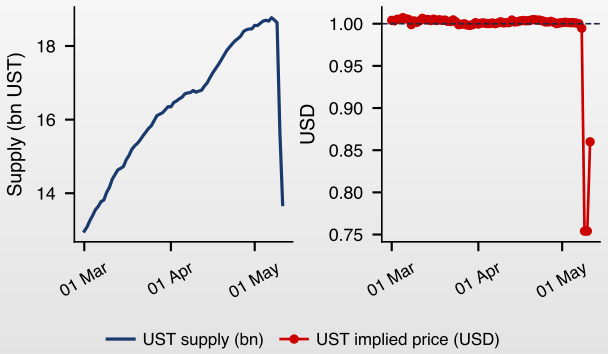
## Arbitrage Bands Imply a Threshold AR: the Hansen Bootstrap

- $p$-value from the fixed-regressor bootstrap of Hansen (1996), valid with heteroskedasticity
  - Artificial series $d_t^* = \hat{\varepsilon}_t \eta_t$: $\hat{\varepsilon}_t$ the residuals of the linear AR(1) (the model under $H_0$), $\eta_t \sim N(0,1)$ independent draws
  - Valid if the scores $d_{t-1}\hat{\varepsilon}_t$ are serially uncorrelated; otherwise $\eta_t$ is held constant on blocks of consecutive days (Shao (2010))
  - The regressors $d_{t-1}$ are held fixed; $\sup_c W^*(c)$ is recomputed on each artificial series; $p$ = share of replications with $\sup_c W^* \geq \sup_c W$

## The Band Width: an Imprecise Estimate

- $\hat{c}$ for USD Coin (2.87 bp) is close to the largest threshold the trimming admits (2.93 bp)
  - the estimate is imprecise
  - Tether: $\hat{c} = 8.57$ bp, largest threshold 8.77 bp; Dai, with 19.6% of the days outside the band, is farther from the boundary
- The half-lives depend on $\hat{c}$
- The linearity test does not: $\sup_c W(c)$ and its bootstrap search the whole grid

# TerraUSD: an Algorithmic Stablecoin Collapses




- Supply and implied price (market value / supply), March–May 2022 (DefiLlama)
- TerraUSD (UST) held its peg by exchange with the token LUNA, not with reserves
  - Demand came from the high yields of the Anchor lending protocol Liu, Makarov & Schoar (2023)
- Supply 18.8 billion UST on 7 May 2022, 13.7 billion four days later (27% redeemed)
  - Implied price 0.754 USD on 10 May 2022
  - The series ends on 11 May 2022; UST fell much further afterwards

MFM_ch16_stablecoins

# Two Stress Episodes: Doubted Backing, Failed Design

## USD Coin, March 2023: the backing

- Tether rose to 1.0077 (high 1.0296): a flight to the other large stablecoin
- Sunday 12 March: all SVB depositors protected Treasury, Federal Reserve & FDIC (2023)
  - The peg returned within two days: the event tested the backing, not the design

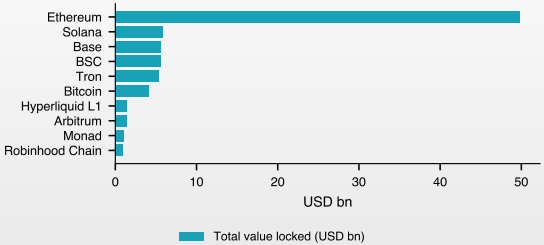
## TerraUSD, May 2022: the design

- Redemption minted more LUNA and pushed its price down: a death spiral
  - within days, UST and LUNA lost almost all their value
  - Sophisticated holders ran first and lost less Liu, Makarov & Schoar (2023)
- Without external collateral the peg rests on confidence alone: a run is self-fulfilling

## Total Value Locked in DeFi

- Monthly $\Delta \ln \mathrm{TVL}_t = a + b\,\Delta \ln P_t^{\mathrm{ETH}} + u_t$, Newey–West SE
  - $b$: the % change in TVL per 1% change in the Ether price $P_t^{\mathrm{ETH}}$: $\hat{b} = 0.79$ (SE 0.07), $R^2 = 0.62$, $n = 79$ complete months (Feb 2020 – Aug 2026)
- TVL mixes token prices with net deposits
  - measure activity with fixed-price quantities, net flows or fees
  - DeFi risks: leverage, liquidity mismatch, interconnection Aramonte, Huang & Schrimpf (2021)

# Where the Value Is Locked




- TVL by blockchain on 18 September 2026 (DefiLlama): the ten largest of the 15 largest at extraction (16 Oct 2026), so a retrospective ranking
  - Ethereum 49.8 billion USD, 57% of 88 billion
  - Solana 5.9, Base 5.6, BSC (BNB Smart Chain) 5.6, Tron 5.4
- Base and Arbitrum are layer-2 networks that settle on Ethereum
  - cheaper transactions, the same security anchor
- L1 in the chart (Hyperliquid L1): a layer-1 network, a blockchain that settles its own transactions

MFM_ch16_defi

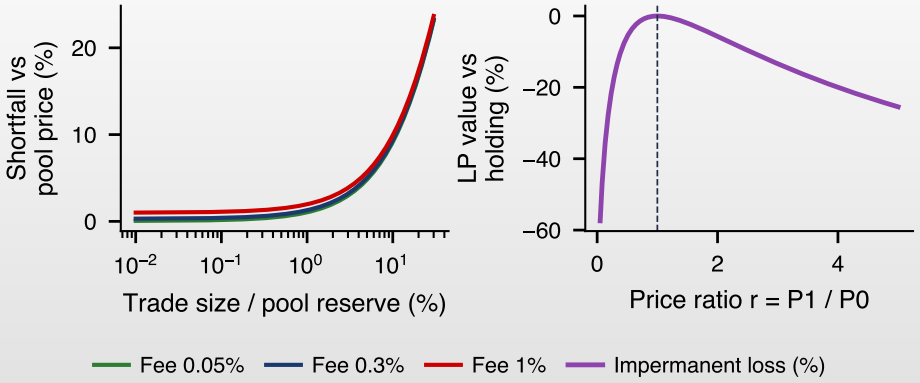
## Worked Example: a Swap in an Ether Pool

- Pool: $x = 2\,000$ Ether, $y = 5\,000\,000$ USD Coin, so $p = 2\,500$ and $k = 10\,000 \times 10^6$; fee $f = 0.3\%$
  - Sell $\Delta x = 20$ Ether ($\delta = 1\%$); after the fee, $\gamma \Delta x = 19.94$
- $\Delta y = \dfrac{5\,000\,000 \times 19.94}{2\,000 + 19.94} = 49\,358$ USD Coin, an execution price of $2\,467.9$ per Ether
- Shortfall: $1 - 2\,467.9/2\,500 = 1.28\%$ (0.3% fee plus about 1% price impact)
- New pool price: $(5\,000\,000 - \Delta y)/2\,020 = 2\,450.8$, i.e. $-1.97\%$
  - Why 2 020 and not 2 019.94: the fee of 0.06 Ether stays in the reserve (Uniswap v2); the product of the reserves rises to $2\,020 \times 4\,950\,642 = 10\,000.3 \times 10^6$, slightly above $k$: the invariant binds for the fee-adjusted input only, and the extra is the liquidity providers' income
  - An arbitrageur now buys Ether in the pool and sells it elsewhere until $p$ is back inside the no-arbitrage band that the fee sets around the market price (Seminar 16, A2)

## The Fee Sets a No-Arbitrage Band

- $P$: the Ether price on other exchanges; $p = y/x$: the pool price
- An arbitrageur can sell Ether to the pool or buy Ether from it
- Selling: one Ether brought to the pool receives, at the margin, $\gamma p$ USD Coin
  - the formula for $\Delta y$ with $\Delta x \to 0$
  - it pays if $\gamma p > P$, i.e. $p > P/\gamma$
- Buying: one Ether taken from the pool costs, at the margin, $p/\gamma$ USD Coin
  - it pays if $p/\gamma < P$, i.e. $p < \gamma P$
- No-arbitrage band: $\gamma P \leq p \leq P/\gamma$, of relative width $1/\gamma - \gamma \approx 2f$
  - Inside the band arbitrage does not cover the fee, and the pool price is not corrected
- In the previous example: $p = 2\,450.8 < \gamma P = 0.997 \times 2\,500 = 2\,492.5$
  - so arbitrageurs buy Ether until the effective pool price reaches $\gamma P = 2\,492.5$, not $2\,500$
  - the fee retained in the reserve takes the reserve ratio to $2\,492.56$, just inside the band

# Slippage and Impermanent Loss (1/2)




MFM_ch16_amm

## Slippage and Impermanent Loss (2/2)

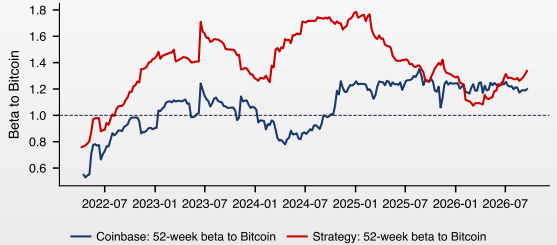
- Left: shortfall by trade size and fee
  - with a 0.3% fee 0.40% for 0.1% of the reserve, 1.28% for 1%, 9.34% for 10%
  - 0.05% fee: 1.04% at 1%
- Small trades pay the fee, large ones the price impact (about $\delta$)
  - liquidity is the depth of the reserve, not an order book (Chapter 10)
  - concentrated liquidity deepens the pool near the price (Uniswap v3)
- Right: value of a liquidity provider (LP) position relative to holding the two tokens
  - against the price ratio $r = P_1/P_0$

MFM_ch16_amm

## The Dimson Beta: Leads and Lags

- Dimson (1979): $r_t^{\text{ETF}} = a + \sum_{j=-1}^{1} b_j \, r_{t+j}^{\text{coin}} + u_t$
  - $b_j$: response to the coin's return on day $t + j$ ($j = -1$: the day before, $+1$: the day after)
  - Sum beta $\hat{\beta}_D = \hat{b}_{-1} + \hat{b}_0 + \hat{b}_1$ collects the whole response
- Which lag should carry the delay?
  - the coin's daily close comes after the 16:00 New York close
  - so the ETF's day-$t$ return already contains the last hours of the coin's day $t - 1$
  - expect $b_{-1} > 0$ and $b_1 \approx 0$
  - IBIT: $\hat{b}_{-1} = 0.05$, $\hat{b}_0 = 0.94$, $\hat{b}_1 = 0.01$, the pattern of a coin close after New York
- IBIT: $\hat{\beta}_D = 1.00$ (SE 0.023) vs OLS 0.94 on the same 671 days
  - ETHA: 0.99 (SE 0.022) vs 0.94: within one SE of 1, as the weekly betas
  - Where prices form is a separate question Hasbrouck (1995)

# Beta to Bitcoin Over Time




- Rolling 52-week beta of weekly returns on Bitcoin; Strategy's ranged from 0.76 to 1.79
  - both stocks are more volatile than Bitcoin (55% a year on the same weeks)
- When did the beta change? A break test at an unknown date: Seminar 16, B3, B4

MFM_ch16_equities

# Testing a Factor Model: Fama–MacBeth

- Premium of a factor: Fama–MacBeth (FM), two passes
  - Pass 1: the betas $\hat{\beta}_i$ from the time-series regressions
  - Pass 2: each week, regress the $N$ excess returns on the $\hat{\beta}_i$ across assets; the slope $\hat{\lambda}_t$ is that week's reward per unit of beta
  - $\hat{\lambda}$ = mean of the $\hat{\lambda}_t$: the factor premium; $\mathrm{SE}_{FM} = \mathrm{sd}(\hat{\lambda}_t)/\sqrt{T}$
- The FM standard error ignores that $\hat{\beta}_i$ is estimated: Shanken (1992) correction
  - Weak factors (small spread of $\beta$ across assets) over-reject Kleibergen (2009)
- GRS judges the model, FM the premium
  - a priced factor in an incomplete model can fail GRS
  - formulas and robust versions: Appendix

## A Small Replication: Features and Method

- 26 assets in two windows
  - 8 cryptos, 6 equity indices (including BET, the main index of the Bucharest Stock Exchange)
  - 4 exchange rates, 4 commodities, 4 bond ETFs
  - W1: January 2019 – June 2021; W2: 11 January 2024 – 18 September 2026 (after the spot ETFs)
- Eight features of daily log returns, each asset on its own calendar
  - Moments: $\ln \sigma_a$ (log annualised volatility), skewness $S$, $\ln(1 + K)$ ($K$: excess kurtosis)
  - Memory: $\hat{\rho}_1(r)$, $\hat{\rho}_1(r^2)$, lag-1 autocorrelations of the returns and of the squared returns
  - Tails: Hill indices of losses and gains (5% largest); maximum drawdown
- **Principal component analysis** (PCA) of the standardised features
  - directions of maximum variance
  - Chapter 3
  - Distance between class centroids in the full 8-dimensional standardised space

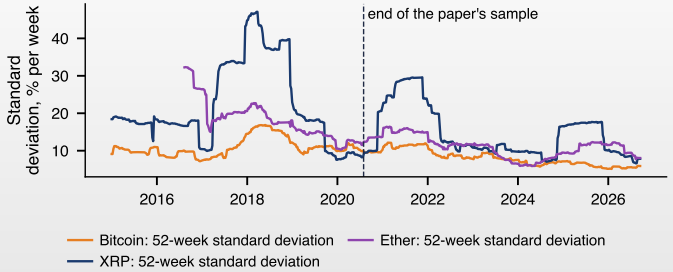
# Is the Move Significant? (1/2): a Block Bootstrap of the Map

- In each window, draw the same calendar days for all 26 assets
  - moving blocks of 28 days: four weeks, 20 trading days
  - Each asset keeps its own returns on those days, so cross-asset dependence is preserved
  - Recompute the eight features, the common standardisation and the centroid distances
  - 500 replications
  - Statistic: change $\Delta = d_{W2} - d_{W1}$ in the distance between the crypto centroid and each class centroid

## Is the Move Significant? (2/2): the Intervals

- Equities: $\Delta = -0.39$, 95% CI $[-1.04, 0.59]$; commodities: $-0.41$ $[-1.38, 0.47]$; both contain 0
  - Currencies: $-0.89$ $[-1.67, 0.18]$; bonds: $-0.62$ $[-1.55, 0.92]$
  - Spread of the cryptos around their centre: $-0.70$ $[-1.09, 0.31]$
- Reading: all five indicative intervals contain 0
  - the map shows cryptos apart
  - it does not establish a reliable post-ETF move towards the other classes
- Caveat: $\ln(1+K)$ and $\hat{\rho}_1(r^2)$ have population targets only if $Er^4 < \infty$
  - doubtful here: they enter as sample descriptors
  - so the map is descriptive and these intervals are indicative

# Case Study: Volatility Before and After the Paper's Sample




- Standard deviation of weekly returns over the last 52 weeks
  - dashed: end of July 2020, the end of the paper's sample
  - peaks: XRP 47.1% (Mar 2018), Bitcoin 16.9% (May 2018) a week
- Last window: Bitcoin 5.9%, XRP 7.8%, Ether 8.0% a week; volatility moves in regimes
  - factor tests across 2020 need HAC errors and a break test at an unknown date
  - Seminar 16, C3

MFM_ch16_ltw

# Replicating Crypto Factors

- **Replicate** with the course data
  - daily closes of the nine non-stable coins (Bitcoin, Ether, XRP, BNB, Cardano, Solana, Dogecoin, Litecoin, Chainlink)
  - one-month Treasury bill rate from the Kenneth French Data Library, the bill of the paper's CMKT
  - Data limit: CSMB and CMOM need breakpoints over hundreds of coins with market values, dead coins included; the nearest published design is the top-20 tercile of Table IA.13, but our nine coins are today's survivors

# Tail Index, Inference Conditions (1/3)

- Regularly varying tail: $P(L > x) = x^{-\alpha}\ell(x)$, with $\ell$ slowly varying
  - slowly varying: $\ell(tx)/\ell(x) \to 1$ as $x \to \infty$, for every $t > 0$
  - the exact power law $Cx^{-\alpha}$ up to a slowly changing factor
  - $n$: number of losses; $k \to \infty$ and $k/n \to 0$: more tail observations, but a vanishing share of the sample
  - Second-order condition $\sqrt{k}\,A(n/k) \to 0$: $A(\cdot)$ measures how fast the tail approaches an exact power law; the condition makes the bias negligible
- Then $\sqrt{k}(\hat{\alpha}_H - \alpha) \Rightarrow N(0, \alpha^2)$, which gives $\mathrm{SE} = \hat{\alpha}_H/\sqrt{k}$
  - The 5% share is a finite-sample tuning choice, not implied by the theory

Back

# Tail Index, Inference Conditions (2/3)

- With volatility clustering the variance depends on extremal dependence Drees (2000)
  - it is typically larger
  - a block bootstrap needs further tail and block-length conditions
  - Choose $k$ from a Hill plot, $\hat{\alpha}_H$ against $k$, in a region where the curve is flat Drees, de Haan & Resnick (2000)
  - Bitcoin: $\hat{\alpha}_H = 2.73$ ($k = 78$), SE 0.31; with $k = 2.5\%$ or $10\%$: 3.48 and 2.59: the choice of $k$ matters as much as the sampling error

Back

# Tail Index, Inference Conditions (3/3)

- Block bootstrap intervals: up to 4.48 for the S&P 500
  - bootstrap CIs for the sample kurtosis are invalid when $\alpha < 4$
  - bootstrap failure with infinite variance, Athreya (1987)
  - Dogecoin: sample kurtosis 106.4, driven by a few social-media jumps, with $\hat{\alpha}_H = 2.83$: it estimates nothing stable
- Equal tails, $H_0$: $\alpha_1 = \alpha_2$: the difference $\hat{\alpha}_1 - \hat{\alpha}_2$, with a joint block bootstrap of 28 calendar days
  - both assets take their returns from the same days, each on its own calendar: common shocks are kept
  - Bitcoin vs S&P 500: $-0.14$, 95% CI $[-1.68, 0.71]$
  - Dogecoin vs S&P 500: $-0.04$, 95% CI $[-1.77, 0.85]$
  - Bitcoin vs gold: $-0.02$, 95% CI $[-1.13, 0.81]$
  - with $k$ at 2.5% or 10% of the losses, all three intervals still contain 0
- Overlapping CIs are not a test of equality, and non-rejection is not equivalence
  - the intervals above admit differences above 1 in $\alpha$: the closeness of the estimates stays a descriptive observation

Back

# The sup-Wald Test, Null Limit and Critical Values

- Its limit is $\sup_r \|B(r) - rB(1)\|^2/[r(1-r)]$
  - $r = \tau/T$: the break fraction
  - $B$: a standard Brownian motion with one component per parameter
  - The sup runs over $r \in [0.15, 0.85]$, the trimming of the main slide; $B(r) - rB(1)$ is a Brownian bridge, zero at $r = 0$ and $r = 1$
  - Critical values simulated from this limit: 11.55 at 5%, 15.18 at 1%, independent of $T$
- Why the trimming: near $r = 0$ or $r = 1$ one regime has very few days
  - $r(1-r) \to 0$ and the supremum over all of $(0, 1)$ would be infinite
- Why the maximum of $W$ does not date the break
  - $W(\tau)$ uses a separate HAC covariance in each regime
  - so its maximiser (here 14 Aug 2017) need not minimise the sum of squared residuals
  - The break is dated by the least-squares estimate (9 Mar 2020) and its interval

Back

# The Published CRIX Rule in Detail

- Total market index TMI (all coins) and candidate indices CRIX$(k, \beta)$ (Trimborn & Härdle (2018), Sect. 3)
  - the $k_1$ largest coins at weight 1
  - the next $s = k - k_1$ with free weights $\beta$
  - $\hat{\varepsilon}_t(k, \beta)$: the tracking residual of the main slide; $\hat{\beta}$ minimises $\|\hat{\varepsilon}(k, \beta)\|^2$, the sum of squared tracking residuals
- $\mathrm{AIC}(k)$ of the main slide
  - $L$ evaluates the residuals under a kernel density estimated from the base residuals $\hat{\varepsilon}(k_1, 1)$
  - Epanechnikov kernel, a parabola $K(u) = \frac{3}{4\sqrt{5}}(1 - u^2/5)$ for $|u| \leq \sqrt{5}$ (variance 1); Sheather–Jones plug-in bandwidth, a data-driven choice of the smoothing width
  - Under Normal errors it reduces to $T \ln \hat{\sigma}_k^2 + 2s$, with $T$ days and $\hat{\sigma}_k^2$ the residual variance (Trimborn & Härdle (2018, author version))
- ECRIX: $k_1 = 1$, steps of one coin, stop at the first rise of the AIC (Trimborn & Härdle (2018, author version), Sect. 5)
  - EFCRIX: global minimum
- The course implementation is a teaching adaptation of eqs. (19)–(20) of Trimborn & Härdle (2018, author version)
  - $\hat{\beta}$ by least squares, then $L$ evaluated at $\hat{\beta}$: not necessarily the maximum of $L$
  - the search stops at 4 of the 5 coins of the universe
  - if the Sheather–Jones equation has no root, the bandwidth falls back to the Normal reference rule $1.06\, \hat{s}\, n^{-1/5}$

Back

# Peg Deviations, Summary Statistics

| | s.d. (bp) | $\lvert d_t \rvert > 50$ (%) | lowest close (bp) | lowest low (bp) |
|---|---|---|---|---|
| Tether | 7.3 | 0.14 | −41 | −515 |
| USD Coin | 7.5 | 0.14 | −285 | −1 226 |
| Dai | 10.8 | 0.57 | −261 | −1 030 |

- Daily close, 1 January 2021 – 18 September 2026
  - $d_t = 10^4(P_t - 1)$ in basis points
  - lowest low from the daily low price

Back

# Tokenisation of Real-World Assets

- **Tokenised yield-bearing instruments**: claims on Treasury bills and cash
  - issued as tokens on a blockchain; the legal form differs
  - Fund shares: BlackRock `BUIDL` 2.38 billion USD (18 Sep 2026, DefiLlama)
  - Circle `USYC` is another tokenised money-market fund share
  - A tokenised secured note issued by a special-purpose vehicle: Ondo `USDY`
- Unlike a payment stablecoin, they pay the Treasury yield to the holder
- Benefits and risks Aramonte, Huang & Schrimpf (2021)
  - benefits: settlement 24/7, use as collateral in DeFi
  - risks: smart-contract bugs, legal ownership, liquidity in stress
- Token economics and adoption: Cong, Li & Wang (2021)

Back

# Decentralised Finance in Five Layers

- Schär (2021) describes DeFi as a stack of layers
  - **Settlement**: the blockchain and its native coin (Ether)
  - **Assets**: tokens issued on it (stablecoins, governance tokens)
  - **Protocols**: exchanges, lending, derivatives as smart contracts
  - **Applications** and **aggregators**: user interfaces that route trades across protocols
- Promise: open access, transparency, composability (protocols plug into each other)
- Risks: high leverage, liquidity mismatches, built-in interconnectedness, no shock absorber Aramonte, Huang & Schrimpf (2021)
  - "Decentralisation illusion": governance tends to concentrate power Aramonte, Huang & Schrimpf (2021)

Back

# Deriving LVR $= \sigma^2/8$ (1/2)

- Pool value at market price $P$ after arbitrage: $V(P) = Px + y = 2\sqrt{kP}$
  - concave: $V'(P) = \sqrt{k/P} = x$, $V''(P) = -\frac{1}{2}\sqrt{k}\,P^{-3/2}$
  - $V'(P)$ is the pool's Ether holding, $V'' < 0$ its negative gamma
  - Rebalancing portfolio, value $R_t$: holds the same $x_t = V'(P_t)$ Ether at every instant, so $dR_t = V'(P_t)\,dP_t$
- Price: geometric Brownian motion $dP_t = \mu P_t\,dt + \sigma P_t\,dW_t$
  - Itô's lemma: $dV_t = V'\,dP_t + \frac{1}{2}V''\,\sigma^2 P_t^2\,dt$
  - $\mu$: drift; $\sigma$: volatility; $W_t$: standard Brownian motion; $V_t = V(P_t)$
  - Subtracting: $d(R_t - V_t) = -\frac{1}{2}V''(P_t)\,\sigma^2 P_t^2\,dt = \frac{1}{4}\sqrt{kP_t}\,\sigma^2\,dt = \dfrac{\sigma^2}{8}\,V_t\,dt$

Back

# Deriving LVR $= \sigma^2/8$ (2/2): Reading and Assumptions

- LVR is the pool's negative gamma Milionis et al. (2022)
  - a short-volatility position that loses $\sigma^2/8$ of its value per unit of time, whatever the drift
- Assumptions: continuous geometric Brownian motion, frictionless arbitrage, no fees, continuous blocks
  - With fees and discrete blocks, arbitrageurs trade only when the price leaves the fee band, which lowers the loss Milionis, Moallemi & Roughgarden (2025)

Back

# The Liquidity-Provider Benchmarks in Detail

- Realised LVR rate $\frac{1}{Y}\sum_t(\Delta R_t - \Delta V_t)/V_{t-1}$ per year, $Y = 2.7$ years; cumulative LVR 17.0 points over 2.7 years
  - $\Delta R_t$, $\Delta V_t$: daily changes in the values of the rebalancing portfolio and of the pool; $V_{t-1}$: the pool value the day before
- The match with theory is not a test
  - each day's loss is $g_t/2 - (\sqrt{1+g_t} - 1) \approx g_t^2/8$
  - $g_t$: the daily simple return of Ether
  - so the realised LVR is approximately the realised variance over 8: 5.83% against $\sum_t g_t^2/(8Y) = 5.88\%$
- 2.7 years pin the realised rate down only to about $\pm 1$ point: CI $[4.88, 6.98]\%$
- The constant 50/50 mix (final value 123.3) is a different benchmark
  - same instantaneous rate: its log growth exceeds the pool's by $\sigma^2/8$, here 5.79% a year

Back

# Crypto Factors, the Literature

- Value from on-chain adoption and segmentation from traditional markets Cong, Karolyi, Tang & Zhao (2026)
  - mispricing versus risk compensation Babiak & Bianchi (2025)
- Fundamentals: a generalized Metcalfe's law describes the long-run value Wheatley et al. (2019), Pele & Mazurencu-Marinescu-Pele (2019a)
  - network value grows with a power of the number of users
  - bubbles are deviations from it
  - adoption and token value reinforce each other Cong, Li & Wang (2021)
- Two different questions, tested separately on the next slides
  - Model adequacy: are all test-asset alphas zero for the chosen factors? (time series, GRS)
  - Risk premium: is a factor's premium nonzero? (a traded factor: its mean excess return; cross-section: Fama–MacBeth $\lambda$)

Back

# Time-Series Test, GRS

- Model, $H_0$: $\alpha_1 = \cdots = \alpha_N = 0$ and the GRS statistic Gibbons, Ross & Shanken (1989): main slide *Testing a Factor Model (2/2)*
  - $\hat{\Sigma}$, $\hat{\Omega}$: residual and factor covariances, both divided by $T$
- The $F$ distribution is exact only for i.i.d. errors with the Normal distribution
  - conditional on the factors
  - crypto returns have $\hat{\alpha}_H$ near 3 and volatility clustering
  - so these assumptions are doubtful for the residuals
  - Robust alternatives: a GMM (generalized method of moments) Wald test with HAC covariance, or a wild bootstrap under $H_0$: $r_t^{e*} = \hat{\beta}' f_t + \hat{\varepsilon}_t \eta_t$
  - $r_t^{e*}$: artificial excess returns with zero alphas; $\hat{\varepsilon}_t$: the residual vector of week $t$; $\eta_t = \pm 1$ with probability 1/2 (Rademacher), one draw for all assets (keeps the cross-correlation)
  - Valid if the residuals are serially uncorrelated (martingale differences), otherwise a block or dependent wild bootstrap
- $\hat{\Sigma}$ must be invertible and precise: $N$ much smaller than $T$; weekly crypto data since 2018 give $T \approx 450$

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# Cross-Section (1/2), Fama–MacBeth and Shanken

- Two passes: $\hat{\beta}_i$ from time series; then each week $r^e_{it} = \gamma_{\mathbf{0}t} + \hat{\beta}_i' \lambda_t + e_{it}$ across the $N$ assets
  - $\gamma_{\mathbf{0}t}$: intercept, the excess return of a zero-beta asset; $\lambda_t$: the reward per unit of beta in week $t$; $e_{it}$: pricing error
  - $\hat{\lambda}$ and $\mathrm{SE}_{FM}$ as on the main slide; $\mathrm{SE}_{FM}$ is too small because $\hat{\beta}_i$ is estimated (errors in variables)
- Shanken (1992), for $\hat{\theta} = (\hat{\gamma}_{\mathbf{0}}, \hat{\lambda}')'$ and $X = [1, \hat{\beta}]$: $\widehat{\mathrm{Var}}(\hat{\theta}) = \frac{1}{T}\left[(X'X)^{-\mathbf{1}} X' \hat{\mathbf{\Sigma}} X (X'X)^{-\mathbf{1}} (1 + \hat{\lambda}' \hat{\mathbf{\Sigma}}_f^{-\mathbf{1}} \hat{\lambda}) + \hat{\mathbf{\Sigma}}_f^*\right]$
  - 1: a column of ones; $\hat{\beta}$: the $N \times K$ matrix of betas; $\hat{\mathbf{\Sigma}}$: residual covariance; $\hat{\mathbf{\Sigma}}_f$: factor covariance; $\hat{\mathbf{\Sigma}}_f^* = \begin{pmatrix} 0 & 0 \\ 0 & \hat{\mathbf{\Sigma}}_f \end{pmatrix}$
  - $\widehat{\mathrm{Var}}(\hat{\lambda})$ is its lower $K \times K$ block; the residual part is multiplied by $1 + \hat{\lambda}' \hat{\mathbf{\Sigma}}_f^{-\mathbf{1}} \hat{\lambda} > 1$, the factor part $\hat{\mathbf{\Sigma}}_f^*$ is added once
  - Assumptions: residuals serially uncorrelated and homoskedastic conditional on the factors

Back

# Cross-Section (2/2), Weak Factors and Multiple Testing

- Weak or useless factors: $\lambda$ is weakly identified and FM $t$-statistics over-reject
  - small spread of $\beta$, e.g. attention proxies
  - Use identification-robust tests of risk premia Kleibergen (2009)
- Multiple testing: dozens of crypto "anomalies" have been tried
  - a new factor should clear $|t| > 3$ Harvey, Liu & Zhu (2016), or control the false discovery rate
  - False discovery rate: the expected share of false findings among the rejected hypotheses
  - In the case study, Liu, Tsyvinski & Wu (2022): $p$-values adjusted under a generalised familywise error rate (k-FWER, probability of $k$ or more false rejections, Lehmann & Romano (2005)), weaker than Holm's FWER (probability of at least one false rejection)

Back

# Data-Quality Threats to Identification

- **Wash trading**: fake volume on unregulated exchanges inflates volume- and liquidity-based factors Cong, Li, Tang & Yang (2023)
  - Filter the exchanges and volumes using the rules of the cited paper before building any factor
- **Which price?** Large, recurrent gaps between exchanges, much larger across countries than within them Makarov & Schoar (2020)
- **Survivorship**: today's large coins are the winners Brown, Goetzmann, Ibbotson & Ross (1992), Będowska-Sójka, Wójcik & Pele (2026)
  - dead coins are missing from most samples
- Remedy: fix the investable universe ex ante (exchange list, volume filter)
  - keep delisted coins, report results by sub-universe

Back

# The Asset-Class Map, PC1 Loadings

| Feature (standardised) | PC1 loading |
|---|---|
| $\ln \sigma_a$, log annualised volatility | 0.44 |
| $\ln(1+K)$, kurtosis | 0.52 |
| Hill index of losses | −0.48 |
| Hill index of gains | −0.34 |
| Maximum drawdown (negative number) | −0.42 |
| Skewness $S$ | 0.14 |
| $\hat{\rho}_1(r^2)$ | 0.08 |
| $\hat{\rho}_1(r)$ | −0.05 |

- A loading is the weight of a standardised feature in PC1
  - the eight squared loadings sum to 1
  - the sign is chosen so that PC1 rises with volatility
  - High PC1: high volatility and kurtosis, low Hill indices (heavy tails), a very negative maximum drawdown (deep falls); skewness and the two autocorrelations hardly enter

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# Common Pitfalls

- Reporting sample kurtosis as a population estimate without discussing whether the fourth moment exists
  - if $\alpha < 4$, the population kurtosis does not exist and its bootstrap CI is invalid
  - $\hat{\alpha}_H < 4$ points to this case, it does not prove it: the interval for $\alpha$ decides
- Testing a threshold or a break date with $\chi^2$ critical values: the nuisance parameter is not identified under $H_0$
- Comparing daily closes recorded at different times and concluding that an ETF tracks badly
- Reading TVL growth as adoption without separating the valuation of the tokens from net deposits
- Treating a stablecoin as risk-free cash: the peg is only as good as redemption and reserves
- Total instead of circulating supply in a market-value index
  - survivorship: studying only today's large coins ignores the many that failed Brown, Goetzmann, Ibbotson & Ross (1992), Będowska-Sójka, Wójcik & Pele (2026)

Back

# The Discovery Loop, Reusable Prompts (1/2)

- **Literature**: map what is known, then verify every reference
  - "`List 2023-2026 papers that test for a change in Bitcoin-equity co-movement at the US spot ETF approval: DOI, sample, method, sign of the effect.`"
  - Resolve each DOI on Crossref, open the paper, find the number it is cited for; search with Semantic Scholar, Elicit or Google Scholar
- **Hypotheses**: competing mechanisms, each with an implication the others do not share
  - "`Give three competing mechanisms for the post-2020 Bitcoin-S&P 500 correlation and, for each, one testable implication that separates it from the others.`"
  - E.g. ETF flows predict an effect concentrated in US trading hours; a common monetary factor predicts co-movement with rate surprises already before 2024
- **Data and code**: on the course data, replicating a known number first
  - "`Take the daily closes of Bitcoin and the S&P 500 from the course data, align the prices on common trading days, then compute log returns and their 2020-2023 correlation.`"
  - Replicate first: this chapter reports 0.39 (2020–2023) and 0.38 (since the ETFs); no new result before the code hits both

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# The Discovery Loop, Reusable Prompts (2/2)

- **Robustness**, fixed before the post-ETF results are seen
  - "`Draft a pre-registered robustness plan: break-date scan, placebo launch dates, weekly returns, tail dependence, block bootstrap; fix every tuning choice now.`"
  - The plan fixes windows, block length and the family of tests: the number of specifications is known and the FWER can be controlled
- **Critique**: ask the assistant to attack the result
  - "`Act as a referee: give the three strongest reasons this change in correlation could be spurious, and the test that would settle each.`"
  - Expect: non-synchronous closes, correlation that rises mechanically with volatility Forbes & Rigobon (2002), confounding events
- **Report**: effect size with a CI, every specification tried, what failed
  - The prompts that shaped the design go into `AI_USE.md`

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# Further Study on Quantinar

## Advanced courses on quantinar.com

- Are Cryptos becoming alternative Assets?
- Introduction to Blockchain and Cryptocurrencies
- Comparison of Cryptocurrency Indices
- Cryptocurrency as an Asset Class
- Smart Contracts
- Stabilising Onchain Stablecoin Risks
- Risk Premia in the Bitcoin Market

## Code and data

- Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_16

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# Acronyms used in this material (1/3)

- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model); in event studies: Abnormal Return
- **BET**: Bucharest Exchange Trading (index)
- **BNB**: BNB (native token of BNB Chain, formerly Binance Coin)
- **BSC**: BNB Smart Chain
- **BTC**: Bitcoin (ticker)
- **CAPM**: Capital Asset Pricing Model
- **CI**: Confidence Interval
- **CMKT**: Crypto Market factor (value-weighted market excess return)
- **CMOM**: Crypto Momentum factor (past winners minus losers)
- **COVID-19**: COronaVIrus Disease 2019
- **CRIX**: CRypto IndeX
- **CSMB**: Crypto Size factor (small minus big coins)
- **DAI**: Dai (stablecoin of the Maker/Sky protocol)
- **DeFi**: Decentralised Finance
- **DOI**: Digital Object Identifier
- **DTB4WK**: 4-Week Treasury Bill Secondary Market Rate (FRED series code)
- **ECRIX**: Exact CRIX (number of constituents chosen in steps of one)
- **EFCRIX**: Exact Full CRIX (global minimum of the AIC)
- **ES**: Expected Shortfall
- **ETF**: Exchange-Traded Fund

# Acronyms used in this material (2/3)

- **ETH**: Ether (ticker of the native coin of Ethereum)
- **ETHA**: iShares Ethereum Trust ETF (ticker)
- **EU**: European Union
- **FAQ**: Frequently Asked Questions
- **FDIC**: Federal Deposit Insurance Corporation
- **FM**: Fama–MacBeth (regressions)
- **FRED**: Federal Reserve Economic Data
- **FWER**: Family-Wise Error Rate
- **FX**: Foreign eXchange
- **GENIUS**: Guiding and Establishing National Innovation for U.S. Stablecoins (Act)
- **GMM**: Generalized Method of Moments
- **GRS**: Gibbons–Ross–Shanken (test)
- **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- **HHI**: Herfindahl–Hirschman Index
- **IA**: Internet Appendix (of the cited paper)
- **IBIT**: iShares Bitcoin Trust ETF (ticker)
- **IL**: Impermanent Loss
- **L1**: Layer 1 (a blockchain that settles its own transactions)
- **LP**: Liquidity Provider
- **LUNA**: Luna (native token of the Terra blockchain)
- **LVR**: Loss-Versus-Rebalancing
- **MAR33**: Basel Framework, chapter MAR33 (internal models approach: capital requirements calculation)
- **MDD**: Maximum DrawDown

# Acronyms used in this material (3/3)

- **MiCA**: Markets in Crypto-Assets (EU Regulation 2023/1114)
- **NAV**: Net Asset Value
- **OLS**: Ordinary Least Squares
- **PC1**: first Principal Component
- **PC2**: second Principal Component
- **PCA**: Principal Component Analysis
- **PYUSD**: PayPal USD (stablecoin)
- **QQQ**: Invesco QQQ Trust, an ETF on the Nasdaq-100 index (ticker)
- **SE**: Standard Error
- **SPX**: S&P 500 index (ticker)
- **SSR**: Sum of Squared Residuals
- **SVB**: Silicon Valley Bank
- **TAR**: Threshold AutoRegression
- **TE**: Tracking Error
- **TMI**: Total Market Index (all coins, CRIX methodology)
- **TVL**: Total Value Locked
- **US**: United States
- **USD**: US dollar, international currency code
- **USDC**: USD Coin (stablecoin issued by Circle)
- **USDS**: Sky Dollar (stablecoin, successor of DAI)
- **USDT**: Tether (stablecoin issued by Tether)
- **UST**: TerraUSD (algorithmic stablecoin of the Terra blockchain)
- **VaR**: Value-at-Risk
- **XAU**: gold, ISO 4217 code (one troy ounce)
- **XRP**: XRP (the native token of the XRP Ledger)

# References (1/6)


- Adams, H., Zinsmeister, N., Robinson, D. (2020). Uniswap v2 Core. Technical white paper.
- Agakishiev, I., Härdle, W.K., Pele, D.T. (2025). Cryptocurrency liquidity forecasting. In *Handbook of Blockchain Analytics*, Springer, 41–60.
- Andrews, D.W.K. (1993). Tests for parameter instability and structural change with unknown change point. *Econometrica*, 61(4), 821–856.
- Angeris, G., Kao, H.-T., Chiang, R., Noyes, C., Chitra, T. (2019). An analysis of Uniswap markets. arXiv:1911.03380.
- Aramonte, S., Huang, W., Schrimpf, A. (2021). DeFi risks and the decentralisation illusion. *BIS Quarterly Review*, December 2021.
- Athreya, K.B. (1987). Bootstrap of the mean in the infinite variance case. *Annals of Statistics*, 15(2), 724–731.
- Babiak, M., Bianchi, D. (2025). Mispricing and risk compensation in cryptocurrency returns. *Journal of Financial and Quantitative Analysis*, 1–27.
- Bai, J. (1997). Estimation of a change point in multiple regression models. *Review of Economics and Statistics*, 79(4), 551–563.
- Bai, J., Perron, P. (1998). Estimating and testing linear models with multiple structural changes. *Econometrica*, 66(1), 47–78.
- Balke, N.S., Fomby, T.B. (1997). Threshold cointegration. *International Economic Review*, 38(3), 627–645.
- Basel Committee on Banking Supervision (2023). MAR33: Internal models approach, capital requirements calculation. Basel Framework, Bank for International Settlements.
- Baur, D.G., Lucey, B.M. (2010). Is gold a hedge or a safe haven? An analysis of stocks, bonds and gold. *Financial Review*, 45(2), 217–229.

# References (2/6)


- Będowska-Sójka, B., Wójcik, P., Pele, D.T. (2026). Early warning systems for cryptocurrency markets: predicting 'zombie' assets using machine learning. *North American Journal of Economics and Finance*, 81, 102543.
- Bitcoin.org (2026). Frequently asked questions. Web page.
- BlackRock (2026). iShares Bitcoin Trust ETF (IBIT). Product page.
- Böhme, R., Christin, N., Edelman, B., Moore, T. (2015). Bitcoin: economics, technology, and governance. *Journal of Economic Perspectives*, 29(2), 213–238.
- Bouri, E., Molnár, P., Azzi, G., Roubaud, D., Hagfors, L.I. (2017). On the hedge and safe haven properties of Bitcoin: is it really more than a diversifier?. *Finance Research Letters*, 20, 192–198.
- Brown, S.J., Goetzmann, W., Ibbotson, R.G., Ross, S.A. (1992). Survivorship bias in performance studies. *Review of Financial Studies*, 5(4), 553–580.
- Buterin, V. (2014). Ethereum: a next-generation smart contract and decentralized application platform. White paper.
- Cong, L.W., Karolyi, G.A., Tang, K., Zhao, W. (2026). Crypto value, factor pricing, and market segmentation. *Management Science*, forthcoming.
- Cong, L.W., Li, X., Tang, K., Yang, Y. (2023). Crypto wash trading. *Management Science*, 69(11), 6427–6454.
- Cong, L.W., Li, Y., Wang, N. (2021). Tokenomics: dynamic adoption and valuation. *Review of Financial Studies*, 34(3), 1105–1155.
- Dimson, E. (1979). Risk measurement when shares are subject to infrequent trading. *Journal of Financial Economics*, 7(2), 197–226.
- Diop, P.O., Chevallier, J., Sanhaji, B. (2024). Collapse of Silicon Valley Bank and USDC depegging: a machine learning experiment. *FinTech*, 3(4), 569–590.
- Drees, H. (2000). Weighted approximations of tail processes for $\beta$-mixing random variables. *Annals of Applied Probability*, 10(4), 1274–1301.
- Drees, H., de Haan, L., Resnick, S. (2000). How to make a Hill plot. *Annals of Statistics*, 28(1), 254–274.

# References (3/6)


- Engle, R.F., Granger, C.W.J. (1987). Co-integration and error correction: representation, estimation, and testing. *Econometrica*, 55(2), 251–276.
- Ethereum.org (2026). The Merge. Web page.
- Fama, E.F., French, K.R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1), 1–22.
- FDIC (2023). FDIC creates a Deposit Insurance National Bank of Santa Clara to protect insured depositors of Silicon Valley Bank. Press release, 10 March 2023.
- Forbes, K.J., Rigobon, R. (2002). No contagion, only interdependence: measuring stock market comovements. *Journal of Finance*, 57(5), 2223–2261.
- Franke, J., Härdle, W.K., Hafner, C.M. (2019). Statistics of Financial Markets: An Introduction, 5th ed. Springer.
- Găman, Ş., Pele, D.T., Härdle, W.K., Będowska-Sójka, B. (2026). White paper on cryptocurrency indices. Bucharest University of Economic Studies.
- GENIUS Act (2025). Guiding and Establishing National Innovation for U.S. Stablecoins Act. Public Law 119-27.
- Gibbons, M.R., Ross, S.A., Shanken, J. (1989). A test of the efficiency of a given portfolio. *Econometrica*, 57(5), 1121–1152.
- Griffin, J.M., Shams, A. (2020). Is Bitcoin really untethered?. *Journal of Finance*, 75(4), 1913–1964.
- Hansen, B.E. (1996). Inference when a nuisance parameter is not identified under the null hypothesis. *Econometrica*, 64(2), 413–430.
- Härdle, W.K., Harvey, C.R., Reule, R.C.G. (2020). Understanding cryptocurrencies. *Journal of Financial Econometrics*, 18(2), 181–208.
- Harvey, C.R., Liu, Y., Zhu, H. (2016). . . . and the cross-section of expected returns. *Review of Financial Studies*, 29(1), 5–68.

# References (4/6)


- Hasbrouck, J. (1995). One security, many markets: determining the contributions to price discovery. *Journal of Finance*, 50(4), 1175–1199.
- Hill, B.M. (1975). A simple general approach to inference about the tail of a distribution. *Annals of Statistics*, 3(5), 1163–1174.
- Kang, H., Lee, S.-G. (2026). Institutionalization without integration: Bitcoin after the spot ETF. *International Review of Economics & Finance*, 110, 105531.
- Kleibergen, F. (2009). Tests of risk premia in linear factor models. *Journal of Econometrics*, 149(2), 149–173.
- Lee, S., Lee, J. (2025). Shifting dynamics: Bitcoin spot ETF approval and Bitcoin's relationships with financial markets. *Applied Economics Letters*, 1–7.
- Lehmann, E.L., Romano, J.P. (2005). Generalizations of the familywise error rate. *Annals of Statistics*, 33(3), 1138–1154.
- Lin, M.-B., Pele, D.T., Ren, R. (2025). Understanding blockchain technology. In *Handbook of Blockchain Analytics*, Springer, 3–18.
- Liu, J., Makarov, I., Schoar, A. (2023). Anatomy of a run: the Terra Luna crash. NBER Working Paper 31160.
- Liu, Y., Tsyvinski, A. (2021). Risks and returns of cryptocurrency. *Review of Financial Studies*, 34(6), 2689–2727.
- Liu, Y., Tsyvinski, A., Wu, X. (2022). Common risk factors in cryptocurrency. *Journal of Finance*, 77(2), 1133–1177.
- Lyons, R.K., Viswanath-Natraj, G. (2023). What keeps stablecoins stable?. *Journal of International Money and Finance*, 131, 102777.
- Ma, Y., Zeng, Y., Zhang, A.L. (2025). Stablecoin runs and the centralization of arbitrage. NBER Working Paper 33882.

# References (5/6)


- Makarov, I., Schoar, A. (2020). Trading and arbitrage in cryptocurrency markets. *Journal of Financial Economics*, 135(2), 293–319.
- MiCA (2023). Regulation (EU) 2023/1114 of the European Parliament and of the Council on markets in crypto-assets. *Official Journal of the European Union*, L 150, 40–205.
- Milionis, J., Moallemi, C.C., Roughgarden, T. (2025). Automated market making and arbitrage profits in the presence of fees. In *Financial Cryptography and Data Security 2024*, Lecture Notes in Computer Science, Springer, 159–171.
- Milionis, J., Moallemi, C.C., Roughgarden, T., Zhang, A.L. (2022). Automated market making and loss-versus-rebalancing. arXiv:2208.06046.
- Nakamoto, S. (2008). Bitcoin: a peer-to-peer electronic cash system. White paper.
- Newey, W.K., West, K.D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3), 703–708.
- Pagnotta, E.S. (2022). Decentralizing money: Bitcoin prices and blockchain security. *Review of Financial Studies*, 35(2), 866–907.
- Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019a). Metcalfe's law and log-period power laws in the cryptocurrencies market. *Economics*, 13(1), 2019-29.
- Pele, D.T., Mazurencu-Marinescu-Pele, M. (2019b). Using high-frequency entropy to forecast Bitcoin's daily value at risk. *Entropy*, 21(2), 102.
- Pele, D.T., Wesselhöfft, N., Härdle, W.K., Kolossiatis, M., Yatracos, Y.G. (2023). Are cryptos becoming alternative assets?. *European Journal of Finance*, 29(10), 1064–1105.
- Schär, F. (2021). Decentralized finance: on blockchain- and smart contract-based financial markets. *Federal Reserve Bank of St. Louis Review*, 103(2), 153–174.

# References (6/6)


- Shanken, J. (1992). On the estimation of beta-pricing models. *Review of Financial Studies*, 5(1), 1–33.
- Shao, X. (2010). The dependent wild bootstrap. *Journal of the American Statistical Association*, 105(489), 218–235.
- Treasury, Federal Reserve, FDIC (2023). Joint statement by Treasury, Federal Reserve, and FDIC. Press release, 12 March 2023.
- Trimborn, S., Härdle, W.K. (2018). CRIX an index for cryptocurrencies. *Journal of Empirical Finance*, 49, 107–122; author version arXiv:2009.09782.
- Wheatley, S., Sornette, D., Huber, T., Reppen, M., Gantner, R.N. (2019). Are Bitcoin bubbles predictable? Combining a generalized Metcalfe's law and the log-periodic power law singularity model. *Royal Society Open Science*, 6(6), 180538.
- White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4), 817–838.